AI-Powered Digital Marketplace for Upcycling Sri Lanka's Textile Waste.

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Project Proposal Report

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Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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Abstract

During cutting, production, and quality control processes, Sri Lanka's textile and clothing manufacturing industry produces a significant amount of fabric waste. Due to the lack of an organized, transparent, and scalable redistribution system, this trash is now burnt or disposed of in landfills, which contaminates the environment, increases carbon emissions, and results in the loss of potentially valuable reusable resources. Accessing dependable and affordable raw materials is a major barrier for small-scale designers, artisans, and SMEs involved sustainably.

To address these issues, this study suggests an AI-powered digital marketplace that combines four key components:

1. An AI-based computer vision system for automated fabric waste classification and cataloguing
2. An environmental impact prediction module that measures CO2 and water savings from waste reuse.
3. A traceability framework with intelligent recommendation and AI pricing capabilities.
4. A marketplace platform with a predictive waste analytics dashboard.

The system uses Random Forest and XGBoost regression models to estimate pricing and environmental effect, Convolutional Neural Networks (CNNs) with transfer learning for fabric classification, and recommendation algorithms to link waste producers with possible purchasers. FastAPI and Docker are used to deploy the solution as scalable microservices.

A working digital marketplace, R2 >0.85 for prediction models, >85% categorization accuracy for fabric waste, and thorough sustainability metrics for every trash transaction are among the anticipated results. By encouraging waste valorization, generating economic possibilities, and lowering environmental impact through data-driven circular economy practices, the initiative promotes SDGs 12 (Responsible Consumption and Production), 8 (Decent Work and Economic Growth), and 13 (Climate Action).

Keywords: Textile Waste Management, Circular Economy, Computer Vision, Convolutional Neural Networks, Environmental Impact Prediction, Machine Learning, Traceability System, Recommendation System, Digital Marketplace, Sustainability Analytics, Sri Lanka

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1. Introduction

1.1 Background and Context

1.1.1 Real-World Problem

In Sri Lanka's textile and clothing manufacturing industry, cutting, production, and quality control processes generate a substantial quantity of fabric waste. A significant portion of this waste is now burnt or disposed of in landfills due to the absence of a systematic, transparent, and scalable redistribution system. Consequences of this approach include environmental pollution, increased carbon emissions, and the loss of potentially valuable reusable materials. [1, 2].

For small-scale designers, craftspeople, and small-to-medium-sized enterprises (SMEs) engaged in upcycling and sustainable fashion, obtaining reliable and reasonably priced raw materials is a challenge. Their capacity to actively engage in circular economy activities is hampered by the absence of visibility, traceability, and standardized valuation of textile waste. [3].

One of the largest industrial sectors in the world, the textile and apparel industry generates millions of tons of textile waste annually. Most of this material is landfilled, even though it can be recycled, reused, or upcycled. [4, 5]. The clothing industry in Sri Lanka contributes significantly to the country's economy, but it also produces a significant quantity of manufacturing waste. [6].

1.1.2 Who Is Affected and Why It Is Important

Numerous stakeholders, including designers, upcyclers, clothing manufacturers, rural artisans, and the public, are impacted by the seriousness of this issue. In terms of the environment, inappropriate textile waste disposal raises greenhouse gas emissions and landfill usage. Manufacturers lose out on potential revenue streams while designers must spend more on production. Socially marginalized groups have fewer opportunities to earn money because of limited access to raw resources. [7].

Nowadays, most clothing factories use manual textile waste sorting, where workers classify waste materials by visually identifying fabric type, color, and quality. This process is inconsistent, time-consuming, and subjective. [8]. The growth of digital circular economy enterprises is further impeded by the lack of trustworthy waste classification techniques.

1.1.3 Statistical Evidence

The textile industry produces about 92 million tons of waste annually worldwide, and fast fashion trends and rising clothing demand are expected to cause this amount to increase significantly. Studies show that less than 15% of textile waste is recycled; the majority is burned or disposed of in landfills. [9]. Textiles the size of a garbage truck are burned or disposed of in landfills all over the world every second. [2].

Although Sri Lanka's textile and apparel sector employs more than 300,000 people and contributes about 7% of the nation's GDP, it also generates large amounts of waste from unused manufacturing. [10].

1.1.4 Analysis of Existing Solutions and Their Limitations

Most textile waste management systems in use today focus on recycling processes, manual resale channels, or generic e-commerce platforms. Global networks such as textile surplus exchanges enable waste commerce, but they lack sophisticated information, sustainability effect monitoring, and localized adaptation. Because they rely on human pricing and classification, most existing solutions are inefficient and error-prone.

Scholarly research has focused on sustainable fashion supply networks, recycling textile waste, and lifecycle assessment (LCA). While machine learning techniques have been employed in certain studies to predict waste or categorize commodities, they are often stand-alone systems that are not integrated into a whole marketplace context. Furthermore, the majority of recent studies do not address micro-batch purchasing and accessibility for small-scale designers and rural artisans.

Some of the present issues include the lack of automated waste value models, poor traceability of waste origin, insufficient visibility of environmental impact, and the absence of AI-driven fabric classification tailored to Sri Lankan textile waste. Furthermore, present technologies do not offer predictive analytics to help producers track and optimize waste creation.

1.1.5 Proposed Solution Overview

An AI-powered online marketplace is the suggested answer, which intends to make it easier for Sri Lankan apparel companies to redistribute and upcycle their textile waste. The system develops a transparent and effective circular economy platform by integrating web technologies, machine learning, and data analytics.

From a system perspective, examples of inputs include images of textile waste, material attributes (weight, fabric type, quality), and transactional data provided by manufacturers. The processing layer uses artificial intelligence models to classify fabric waste using computer vision techniques, predict resale value using regression models, calculate environmental effect metrics, and generate intelligent recommendations for buyers. The output layer presents pricing estimations, sustainability effect indicators, traceability information, and categorized waste listings via an interactive marketplace dashboard.

By automating categorization and value, allowing micro-batch purchase, and establishing a digital traceability passport for every trash batch, the solution immediately fills the holes found in the current systems. For all parties involved, this guarantees openness, confidence, and data-driven decision-making.

Four primary modules make up the system:

1. AI-driven fabric classification and cataloging module
2. Waste value module and environmental effect analytics.
3. AI pricing module, recommendation system, and traceability
4. Predictive analytics dashboard and marketplace platform

1.1.6 Sustainable Development Goals (SDG) Alignment

The following Sustainable Development Goals are supported by this study by promoting waste valorization and sustainable industrial practices:

- SDG 12 (Responsible Consumption and Production): Encouraging the effective use and repurposing of textile materials while cutting waste through circular economy methods [11].
- SDG 8 (Decent Work and Economic Growth): Providing small-scale designers, artisans, and SMEs with access to reasonably priced raw materials to provide economic opportunities.
- SDG 13 (Climate Action): Cutting greenhouse gas emissions from the production of virgin materials and the disposal of textile waste.
- SDG 9 (Industry, Innovation, and Infrastructure): Using digital technology and predictive analytics to promote sustainable industrial practices.

1.1.7 Research Cluster Alignment

The project, which is part of the Artificial Intelligence research cluster and belongs to the TIM-Technology Integration and Management research group, focuses on the textile and apparel industry vertical to utilize intelligent digital solutions to bridge the gap between waste generation and sustainable reuse.

1. Literature Review

1.2.1 Computer Vision for Textile Classification

Automated material recognition now has more options thanks to recent developments in computer vision and artificial intelligence. Visual patterns, such as color, texture, and structure, can be automatically recognized by computer vision systems. [12]. These systems can learn to identify material attributes with high accuracy by using deep learning models such as Convolutional Neural Networks (CNNs) [13].

Researchers have studied numerous methods for identifying textile materials. By collecting distinct spectrum reflection patterns of materials across several wavelengths, hyperspectral imaging has shown the capacity to distinguish between natural fibers like cotton and synthetic fibers like polyester. [14]. However, the need for costly sensors, regulated illumination, and specialized equipment limits the application of hyperspectral imaging in industrial settings. [15].

The rapid advancement of deep learning technology has made CNNs the state-of-the-art approach for picture categorization challenges. Strong performance in extracting hierarchical visual characteristics from complicated image data has been shown by well-known designs such as ResNet, VGG, and EfficientNet. [16]. Textile-related issues, such as fabric flaw detection and texture classification, have been addressed using CNN-based techniques. [17, 18].

Despite these developments, a major drawback of current research is that the majority of datasets consist of neat, flat, and well-organized fabric samples that were taken in a controlled laboratory setting. [19]. On the other hand, actual textile waste looks asymmetrical, folded, crumpled, and occasionally partially ruined. The requirement for waste-specific datasets is highlighted by the fact that models trained on pristine datasets often perform poorly on images of post-industrial textile waste. [20].

Another crucial method that shortens training times while increasing classification accuracy is transfer learning, which makes it possible to modify models that have already been trained on huge datasets like ImageNet for new uses [21]. Nevertheless, there is little research on using transfer learning techniques specifically for the classification of textile waste in circular economy platforms. [22].

1.2.2 Lifecycle Assessment and Environmental Impact Prediction

A regulated process for measuring a product or material's environmental effects over the course of its whole lifecycle, from the extraction of raw materials to disposal at the end of its useful life, is called Lifecycle Assessment (LCA) [23]. LCA studies have repeatedly shown the substantial environmental impact of virgin fiber manufacturing in the textile sector. For example, 10,000–20,000 liters of water are needed to produce 1 kg of typical cotton, while mechanized farming and fertilizer use significantly increase CO2 emissions. [24]. Despite using less water, the production of polyester uses a lot of energy and fossil fuels, which increases carbon emissions. [25].

The fundamental information, often referred to as "characterization factors," that quantifies impact per unit of material is provided by these investigations. The Ecoinvent database, peer-reviewed life cycle assessment (LCA) publications, reports from organizations like UNEP, and the Higg Materials Sustainability Index (MSI) of the Sustainable Apparel Coalition are examples of authoritative sources. [26].

When garbage is recycled or reused, the environmental benefit of avoided production can be measured using the "avoided burden" or "system expansion" technique in life cycle assessment. [27]. For instance, the effects of making 1 kilogram of virgin cotton are avoided if 1 kg of cotton waste is upcycled. Consequently, the LCA impact of the virgin material it replaces is equal to the environmental benefits (savings of CO2 and water).

Such sustainability data is not included in the textile waste platforms that are already in use. They make garbage transactions easier, but they don't measure the waste's environmental value, which means they lose out on the chance to develop data-driven incentives for sustainable consumption. [28].

1.2.3 Digital Traceability in Supply Chains

The capacity to follow a product's origin, flow, and lifecycle across a supply chain is known as traceability. [29]. All industries employ digital traceability solutions to guarantee regulatory compliance, improve quality control, and increase transparency.

Traceability can offer important details regarding the sources of materials, manufacturing procedures, and distribution routes in the textile sector. Stakeholders can gain a better understanding of the lifecycle of textile materials by documenting and monitoring material properties such as fabric type, quantity, quality grade, and origin. [30].

Digital traceability solutions are crucial for encouraging sustainability and ethical sourcing in supply chains, according to research. Traceability systems enhance sustainable decision-making by allowing businesses and customers to confirm the ethical and environmental characteristics of products. [31]. Nevertheless, the majority of textile waste exchange platforms now in use do not have integrated traceability features that allow waste batches to be systematically tracked throughout their lifecycle.

1.2.4 AI-Based Recommendation and Pricing Systems

Recommendation systems in a variety of fields, such as e-commerce, digital marketing, and resource management, have made extensive use of artificial intelligence. [32]. To provide users with relevant products or actions, recommendation systems examine trends in user behavior, product attributes, and past data.

AI-based recommendation systems can assist in finding appropriate reuse alternatives for items that could otherwise be thrown away in the context of waste management and resource redistribution. Recommendation algorithms can find possible customers, recycling options, or alternative reuse paths by evaluating characteristics like fabric type, quality, and quantity. [33].

Supervised regression methods, such as Random Forest, Gradient Boosting, and Linear Regression, have been effectively used for pricing prediction to anticipate market values based on material attributes and past transaction data. [34]. Data-driven appraisal is made possible by these models' ability to capture intricate correlations between material characteristics and market pricing.

1.2.5 Predictive Analytics for Waste Forecasting

The use of machine learning to forecast industrial production results and waste generation trends has been thoroughly studied by researchers. [35]. By analyzing past data to find trends and project future trash quantities, predictive analytics helps improve recycling and resource management strategy planning.

Predictive models assist firms in waste management by forecasting future trash volumes and creating more efficient recycling plans. Neural networks, regression models, and ensemble learning techniques are examples of machine learning methodologies that have been used to evaluate industrial data and increase prediction accuracy. [36].

However, rather than precisely predicting the amounts of textile waste, the majority of current research focuses on energy consumption or production optimization, which highlights the need for specialist predictive models created for textile waste analysis.

1.2.6 Analytical Reporting and Visualization

Tools for reporting and data visualization are crucial parts of contemporary data-driven systems. [37]. By displaying data via graphs, charts, and visual reports, analytical dashboards assist companies in understanding complicated datasets.

Visualization technologies in waste management systems allow stakeholders to assess recycling effectiveness, track trash creation trends, and comprehend environmental effects. Organizations may swiftly spot patterns and make well-informed decisions about waste management tactics by displaying waste information and forecasting outcomes in dashboards. [38].

1.2.7 Circular Economy and Digital Platforms

As a sustainable substitute for the conventional linear "take-make-dispose" model, the circular economy idea has drawn more and more attention. [39]. The circular economy seeks to prolong the useful life of resources, maximize their value while they are in use, and recover and renew materials and products at the end of each service life.

In the textile industry, digital platforms are becoming more widely acknowledged as useful instruments for enhancing resource circulation and promoting circular economy principles. [40]. These platforms can help make the shift to more sustainable patterns of production and consumption by making waste materials visible and putting trash producers in touch with possible consumers.

1.3 Research Gap

There are several important gaps at the nexus of computer vision, LCA, machine learning, and digital markets for textile waste, according to a thorough analysis of current literature and systems.

1. **Lack of Sri Lanka-Specific Textile Waste Classification Systems**
   The majority of current CNN-based models are intended for fabric quality inspection or defect identification rather than textile waste categorization, even though computer vision approaches for textile analysis have been researched. [41]. Additionally, a few localized datasets represent the textile waste produced by Sri Lankan clothing industries. [42]. On post-industrial trash, which appears folded, crumpled, and occasionally torn, models trained on generic cloth photos perform poorly.

2. **Absence of Batch-Level Environmental Impact Quantification**
   Even though the textile industry has macro-level life cycle assessment (LCA) studies, no approach automatically uses this data to determine the environmental savings of heterogeneous textile waste batches. It is challenging to include environmental value in purchasing choices because most manufacturers, small-scale designers, and craftsmen lack access to LCA expertise, which is highly specialized. [43].

3. **Limited Integration of AI in Textile Waste Platforms**
   Predictive analytics, intelligent suggestions, and environmental effect measurements are not currently included in digital garbage trading systems. These systems lack chances for data-driven incentives for sustainable consumption and only serve as basic listing services. [44]. The business case for waste valorization is weakened by the lack of measurable effect data, which also makes it challenging to monitor advancement toward sustainability objectives.

4. **Lack of Traceability in Waste Redistribution Platforms**
   Users are unable to track the origin, quality, and lifetime of textile waste materials due to the lack of full traceability capabilities on many of the current textile waste exchange

systems. In addition to discouraging sustainable reuse methods, this lack of openness erodes trust between buyers and sellers [45].

5. **Insufficient Integration of Multiple Capabilities**

   Automated categorization, environmental effect measurement, traceability, intelligent suggestions, AI pricing, predictive analytics, and marketplace capabilities particularly for textile waste management, are presently not unified by any system or platform. Because these components usually function independently, stakeholders are unable to use thorough data-driven insights for efficient decision-making.

6. **Limited Decision Support for Waste Utilization**

   Intelligent assistance to assist users in determining the best alternatives for reuse or redistribution is not provided by current technologies. In current digital platforms, recommendation algorithms designed especially for the redistribution of textile waste are still mostly unexplored. [46].

   By creating an integrated AI-powered digital marketplace that integrates computer vision, environmental intelligence, traceability, recommendation systems, and predictive analytics, especially for Sri Lanka's textile waste context, this project seeks to close these gaps.

2. Problem statement

The primary research issue is the lack of an intelligent, integrated digital system for managing textile waste that integrates data-driven decision support, transparent traceability, automated categorization, and environmental effect quantification. This issue can appear in several ways:

- **For the Environment:** Burning or landfilling textile waste improperly results in greenhouse gas emissions, contaminated soil and water, and the loss of natural resources utilized in the manufacturing of virgin fabrics. Both economically and ecologically, the linear "take-make-dispose" cycle is unsustainable. [2]
- **For Manufacturers (Sellers):** They lose out on chances to make money from waste materials since they don't have an automated system to categorize and organize garbage. They are unable to raise potential "green premiums" or enhance ESG reporting by successfully marketing the sustainability benefits of their garbage.
- **Buyers (Designers, Upcyclers, SMEs):** They have trouble locating trustworthy suppliers of reasonably priced raw materials. The environmental value is an abstract idea rather than a concrete, comparative metric since they lack the competence to measure the environmental advantages of selecting recycled materials over virgin textiles.
- **For the Circular Economy:** The creation of transparent, scalable trash exchange networks is hampered by the lack of traceability and consistent value. Stakeholders are unable to validate quality claims, monitor the origin of materials, or make data-driven waste utilization decisions.

Therefore, the main queries that this study aims to address are:

1. How can textile waste (fabric kind, color, quality grade) be accurately and automatically classified from photos using computer vision techniques?
2. How can machine learning algorithms, based on observable attributes, properly forecast the environmental effect (water and CO2 savings) of recycling certain batches of textile waste?
3. In what ways may AI-powered pricing models and recommendation systems help trash producers and future consumers match optimally and transparently?

4. How can proactive planning for recycling and reuse be supported by predictive analytics, forecasting patterns of textile waste generation?
5. How may these features be included in a unified online marketplace that supports circular economy principles?

3. Objectives

3.1 Main Objective

To create and assess a digital marketplace framework driven by artificial intelligence that facilitates intelligent textile waste categorization, valuation, environmental effect analysis, and redistribution in Sri Lanka to promote circular economy principles.

3.2 Specific Objectives

In accordance with the system modules, the research is separated into four main parts.

Module 1: AI Fabric Classification (Sandeepa S.A.C. - IT22281814)

1. To look at the shortcomings of the Sri Lankan clothing industry's existing manual methods for sorting textile waste.
2. To create an image-based dataset of textile waste items with fabric type, colour, and quality grade labels.
3. To use computer vision techniques to create and train a deep learning model for autonomous textile waste categorization.
4. To assess the suggested categorization model's performance using measures like F1-score, accuracy, precision, and recall.

Module 2: Environmental Impact Prediction (Siriwardhana N.S.T.N - IT22215338)

1. To create an organized collection of environmental indicators from Lifecycle Assessment (LCA) for typical textile materials.
2. To create feature engineering methods that translate characteristics of textile waste into input variables for machine learning.
3. To create and train regression models to forecast water savings and a decrease in CO2 emissions from the reuse of textile waste.
4. To use R2, MAE, and RMSE measures to assess the prediction models' performance.
5. To create a Green Score sustainability metric that reflects the advantages of reusing textile waste for the environment.

Module 3: Traceability, Recommendations & AI Pricing (Madushanka H.M.T.G - IT22221278)

1. To create a digital traceability system for monitoring batches of textile waste and the characteristics of their materials.
2. To create an AI-based pricing model that uses past transaction data to forecast the best resale value for textile waste materials.
3. To create an intelligent recommendation system that uses material properties and market demand to determine appropriate reuse or recycling alternatives for textile waste.

Module 4: Marketplace Platform & Predictive Analytics (Thalagala T.A.D.V - IT22281678)

1. To create a digital marketplace platform that makes textile waste materials visible and makes it easier for buyers and producers to redistribute them.
2. To create a predictive analytics model that uses past data to project future amounts of textile waste.
3. To produce visual dashboards and analytical reports to assist data-driven decision-making in the management of textile waste.

4. Methodology

4.1 Overall Research Approach

This study combines experimental machine learning techniques with a design science research methodology. Since design science focuses on developing and assessing novel IT products meant to address recognized organizational issues, it is suitable for this project. [47]. Six steps make up the methodology: problem identification and motivation, solution objectives formulation, design and development, demonstration, evaluation, and communication.

Each team member's module is represented by one of the project's four concurrent development tracks, which are continuously integrated to guarantee system functionality. Data collection, feature engineering, model building, validation, and API deployment are all part of the methodical development process that each module goes through.

4.2 Module 1: AI Fabric Classification (Sandeepa S.A.C. - IT22281814)

4.2.1 Dataset Creation and Augmentation

Data Collection:

Samples of fabric waste will be gathered from textile wholesalers, small apparel makers, and local tailor shops in the Colombo area. Standard smartphone cameras will be used to take pictures of the samples in a variety of lighting, backdrops, and perspectives. By simulating actual user submissions, this method strengthens the classification model's resilience.

To boost dataset variety and enhance generalization capacity, publicly accessible textile datasets will be utilized in addition to locally gathered photos.

Potential additional datasets consist of:

- https://images.cv/dataset/fabric-image-classification-dataset - includes labelled fabric photos that are helpful for deep learning model training.
- https://github.com/hahashu/TextileNet - a large textile dataset with material taxonomy names for hundreds of thousands of fabric photos.
- https://data.mendeley.com/datasets/3vc56ddjhw/2 - includes enlarged samples and pictures of cotton fabric used in textile categorization studies.
- https://huggingface.co/papers/trending - comprises fabric samples that were taken in various lighting situations to replicate real-world settings.

These datasets lessen dataset bias and enhance model generalization.

Data Labeling:

Three properties listed in Table 1 will be manually applied to each photograph. To guarantee labelling uniformity across all samples, a thorough labelling rubric will be developed.

Table 1: Target Classification Categories.

Attribute	Categories	Notes
Fabric Type	Cotton, Polyester, Nylon, Viscose, Blend	"Blend" for common mixtures such as poly-cotton
Color	Red, Blue, Green, Yellow, Black, White, Other	Dominant color; "Other" for complex or multi-colored patterns
Quality Grade	High, Medium, Low	Determined based on visible tears, stains, fading, and structural integrity

Target Dataset Size:

At least 5,000 labelled photos with a balanced distribution across categories will be included in the collection.

This will be accomplished by gathering between 1,500 and 2,000 distinct photos and using data augmentation methods.

Data Augmentation:

To enhance model generalization and avoid overfitting, real-time augmentation will be carried out during model training using Keras ImageDataGenerator.

Techniques for augmentation include:

- Rotation at random (±20°)
- Changes in width and height
- Flipping both horizontally and vertically
- Zoom at random.
- Adjusting brightness

Data Splitting:

Three subsets will be created from the dataset:

Table 2: Data Splitting in Fabric Classification.

Dataset	Percentage	Purpose
Training Set	70%	Model training
Validation Set	15%	Hyperparameter tuning and overfitting monitoring
Test Set	15%	Final performance evaluation

4.2.2 CNN Architecture and Transfer Learning

Base Architecture

The basis model will be a contemporary Convolutional Neural Network (CNN) architecture that has been pre-trained on ImageNet.

Among the potential architectures are:

- ResNet-50
  - Effective training of deep networks is made possible by residual connections.
  - Established architecture for challenges involving the categorization of images.
- EffectiveNetB3
  - Uses fewer parameters to achieve great accuracy.
  - Enhanced computing efficiency through optimized scaling.

Transfer Learning Approach

1. Load the chosen base model that has been pre-trained on ImageNet, excluding the top classification layer.
2. To preserve learnt visual characteristics, freeze the base model's first layers.
3. Include personalized categorization layers:
   - Layer of Global Average Pooling
   - ReLU-activated dense layers.
   - Three levels of output, one for each attribute.

Activation functions:

Table 3: Activation functions.

Attribute	**Activation**
Fabric Type	Softmax
Color	Softmax
Quality Grade	Softmax

The model can concurrently predict fabric type, colour, and quality grade while sharing common visual feature representations thanks to its multi-output design.

4.2.3 Model Training and Evaluation

Environment

- Python is a programming language.
- Framework: Keras/TensorFlow
- Cloud GPU (Google Colab Pro or AWS EC2 GPU instance) is the training hardware.

Procedure for Training

1. Assemble the model using the proper metrics and loss algorithms.
2. Use the training dataset to train the model.
3. To track performance, use the validation dataset.
4. To avoid overfitting, use early halting based on validation loss.
5. You can choose to use a modest learning rate to fine-tune the foundation model by unfreezing its top layers.

Evaluation Metrics

The model's performance will be assessed using:

Table 4: Evaluation Metrics.

Metric	Description
Overall Accuracy	Percentage of images where all attributes are correctly predicted
Per-Attribute Accuracy	Accuracy for fabric type, color, and quality individually
Precision	Correct positive predictions per class
Recall	Ability to correctly detect each class
F1-Score	Harmonic mean of precision and recall

Over 85% overall classification accuracy on the test dataset is the target performance.

4.2.4 API Development

Structure

FastAPI will be used to create the categorization service because of its automated API documentation, asynchronous support, and excellent speed.

Endpoint

POST /classify_fabric

Input:

- Upload an image (multipart/form-data)
- JPG and PNG are supported formats.

Processing Pipeline

1. Verify and prepare the picture.
2. Resize to the model input scale (224 x 224 pixels, for example).
3. Load the CNN model that has been trained.
4. Execute inference.
5. Make forecasts using confidence levels.

Example API Response:

```
{
  "fabric_type": "Cotton",
  "color": "Blue",
  "quality_grade": "Medium",
  "confidence_scores": {
    "fabric_type": 0.92,
    "color": 0.88,
    "quality_grade": 0.84
  }
}
```

Figure 1: JSON Type API Response.

4.3 Module 2: Environmental Impact Prediction (Siriwardhana N.S.T.N - IT22215338)

4.3.1 Data Collection and Sourcing

The environmental impact prediction component primarily utilizes secondary data sources, as conducting comprehensive Lifecycle Assessment (LCA) studies is beyond the scope of an undergraduate research project. Instead, it relies on verified environmental impact factors obtained from established databases and reputable academic publications.

Data Sources

The following reliable sources will be used to create the environmental impact dataset:

1. **Ecoinvent**
   One of the most popular life cycle assessment (LCA) databases in the world, with environmental impact information for thousands of products and industrial processes.
   Dataset access:
   https://ecoinvent.org
2. **European Platform on Life Cycle Assessment**
   offers datasets and environmental effect characterization variables for use in sustainability studies.
   https://eplca.jrc.ec.europa.eu
3. **Sustainable Apparel Coalition – Higg Materials Sustainability Index (MSI)**
   Provides industry-validated environmental impact scores for textile materials.
   https://apparelcoalition.org/the-higg-index/
4. Peer-reviewed academic studies on textile lifecycle assessments, such as:

   - **Chapagain et al. (2006)** – Water footprint of cotton production
   - **Shen et al. (2010)** – EnvironmentTextile Waste Attributes: Details provided by vendors at the time of listing (fabric type, quantity, quality grade, color, condition, source)
   - all impacts of polyester production

Dataset Structure

Each row in the assembled dataset, which will be saved in CSV format, will represent a kind of textile material together with its corresponding environmental effect indicators.

Table 5: Dataset Structure Environmental Impact Prediction.

Field	Description
material_name	Specific textile material
material_category	Natural / Synthetic / Semi-synthetic
co2_factor	CO_2 reduction factor (kg CO_2 eq saved per kg reused)
water_factor	Water saving factor (liters per kg reused)
source_reference	Citation of the LCA data source

Table 6: Sample LCA Factors for Common Textile Materials.

Material Name	CO_2 Factor (kg CO_2 eq/kg)	Water Factor (L/kg)
Cotton (Conventional)	5.0	10,000
Polyester (Virgin)	7.5	100
Nylon 6,6	9.0	150
Viscose (Generic)	4.5	600

The sustainability advantages of reusing textile waste will be estimated using these environmental characteristics as baseline indicators.

4.3.2 Feature Engineering

Key Features of Feature Engineering:

- fabric_type (Categorical): The primary key used to connect to the LCA factor database is fabric type.
- weight_kg (Numerical): The total weight of the garbage batch expressed in kilograms.
- High, Medium, and Low serve as the discount factors for quality_grade (Categorical/Ordinal) (e.g., High=1.0, Medium=0.8, Low=0.5).

Target variables and derived features:

- Total_co2_saved (numerical): Co2_factor × weight × quality_multiplier is the target result.
- Total water saved (numerical): Water factor × weight × quality multiplier is the target outcome.
- Green_score (Numerical/Categorical): Based on normalized CO2 and water savings, a composite score (A-F grade or 1-100 scale)

4.3.3 Model Development and Selection

Approach A: Direct Calculation (Lookup Table) - This highly interpretable, rule-based baseline utilizes a lookup table and formula.

Method B: Supervised Regression Models: These machine learning models enable learning from actual transaction data by capturing non-linear interactions.

- Simple baseline model using linear regression.
- An ensemble technique that captures non-linear correlations with feature importance is the Random Forest Regressor.
- Modern outcomes on structured data using the Gradient Boosting Regressor (XGBoost) [48]

By altering LCA factor data across weights and quality grades with little noise additions, training data will be artificially created.

4.3.4 Model Training and Validation

Data splitting involves dividing a synthetic dataset into training (80%) and testing (20%) sets.

Training: To reduce prediction error, regression models are trained.

Metrics for Validation:

- R-squared (R^2): The main indicator of variance explanation, with a goal of >0.85
- The average absolute difference, or mean absolute error (MAE), is better when it is smaller.
- Large errors are given more weight by the Root Mean Square Error (RMSE).

Model Selection: The top-performing model was chosen using interpretability and validation measures.

4.3.5 API Development

Framework: FastAPI for high-performance microservice.

Endpoint: POST /predict_impact

Request Body:

```
{
"fabric_type": "Cotton",
"weight_kg": 5.0,
"quality_grade": "Medium"
}
```

Figure 2: JSON format Request Body.

Response Body:

```
{
"fabric_type": "Cotton",
 "weight_kg": 5.0,
 "quality_grade": "Medium",
  "co2_saved_kg": 20.0,
   "water_saved_l": 40000,
   "green_score": "A",
    "confidence": 0.92
}
```

Figure 3: JSON Format Response Body.

Features include automated Swagger UI documentation and strong error handling for incorrect inputs.

4.4 Module 3: Traceability, Recommendations & AI Pricing (Madushanka H.M.T.G - IT22221278)

4.4.1 Data Collection and Sourcing

Data Sources:

- Textile Waste Attributes: Details provided by vendors at the time of listing (fabric type, quantity, quality grade, color, condition, source)
- Historical Transaction Data: Past sales records with buyer details, selling price, and material characteristics.
- Industry Market Data: Trends and resale values of the publicly accessible textile waste market

Dataset Structure: Every entry corresponds to a batch of textile waste with:

- Type of Fabric
- Quantity/Weight
- Grade of Quality
- Manufacturer/Source
- Condition of Material
- Past Selling Price

Industry Market Data

Publicly available textile resale data and sustainability market insights will be used to understand price ranges and demand trends.

Relevant industry data sources include:

- **Ellen MacArthur Foundation** – textile circular economy reports https://ellenmacarthurfoundation.org
- **Global Fashion Agenda** – sustainability market insights https://globalfashionagenda.org

- Textile recycling market datasets available from **Kaggle** https://www.kaggle.com

These sources provide reference pricing patterns and demand indicators.

4.4.2 Feature Engineering

Raw listing data is transformed into variables that are suitable for machine learning using feature engineering.

Primary Features for Pricing Model

Feature	Type	Description
fabric_type	Categorical	Influences market demand and price
weight_kg	Numerical	Determines batch value
quality_grade	Ordinal	High-quality materials command higher prices
material_condition	Categorical	Physical state of textile waste
source_manufacturer	Categorical	Source identification for traceability
color	Categorical	May influence design use cases

One-Hot Encoding or Label Encoding will be used to encode categorical variables.

Target Variable

- Variable: predicted_resale_price
- Description: Estimated resale value of textile waste batch.

Suggested Features:

- Scores for material similarity
- Buyer preferences in the past
- Demand trends by region and kind of material.

4.4.3 Model Development and Selection

Techniques for continuous price prediction using supervised regression:

- Baseline model for linear regression
- Random Forest Regressor: Non-linear relationship ensemble approach
- Gradient Boosting (XGBoost): A sophisticated method for increased precision

System of Recommendations:

- Content-Based Filtering: Uses attribute matching to suggest related content.
- Collaborative Filtering: Finds trends in past purchasing behavior.
- Hybrid Approach: Integrates both techniques to provide better suggestions.

4.4.4 Traceability Framework

Digital Material Passport Design:

Every trash batch will have a digital passport that includes the following information:

- Distinct Batch ID
- Details about the source (maker, date, location)
- Features of the material (kind, weight, quality grade)
- History of transactions (transfers, sales)
- Metrics for environmental effect (from Module 2)
- Certifications of quality (if relevant)

Traceability Tracking:

- Blockchain-based ledger for unchangeable transaction records
- Each batch generates a QR code that allows stakeholders to access passport data.
- A record of every material movement on the platform

4.4.5 API Development

Framework: FastAPI

Endpoints:

- POST /predict_price - AI pricing predictions.
- POST /get_recommendations - Buyer/reuse recommendations.
- GET /traceability/{batch_id} - Retrieve material passport.
- POST /create_passport - Generate a new material passport.

```
{
 "fabric_type": "Polyester",
 "weight_kg": 10,
 "quality_grade": "High",
 "material_condition": "Roll Ends"
}
```

Figure 4: JASON Format Response Body

```
{
 "predicted_resale_price": 48.50,
 "recommended_buyers": ["Upcycling Studio A", "Craft Collective"],
 "confidence": 0.91
}
```

Figure 5: JASON Format Response Body

4.5 Module 4: Marketplace Platform & Predictive Analytics (Thalagala T.A.D.V - IT22281678)

4.5.1 Data Collection and Sourcing

To assist sustainable reuse and recycling options, the marketplace analytics module focuses on gathering data on textile waste, examining past patterns, and projecting future trash availability.

Data Sources

Several sources will be used to generate the dataset:

1. **Industry Waste Data**

Data on textile waste will be gathered from:

- Manufacturing facilities for clothing
- Small and medium-sized textile businesses (SMEs)
- Organizations that recycle and upcycle

This data will consist of:

- Types of textile materials
- Quantities of waste produced.
- Dates of waste availability
- Geographical places

2. **Historical Waste Records**

To find patterns and trends in the creation of textile waste, historical waste generation information from partner organizations will be utilized.

These documents might consist of:

- Quantities of garbage generated per month.
- Statistics on the distribution of materials.
- Records of waste reuse and recycling.

3. **Public Benchmark Datasets**

To improve model reliability and benchmarking, publicly available textile datasets may also be used.

Examples include:

- Kaggle Textile Waste Dataset
  https://www.kaggle.com/datasets
- Global textile waste statistics from the **United Nations Environment Program**
  https://www.unep.org
- Circular economy textile reports from **Ellen MacArthur Foundation**
  https://ellenmacarthurfoundation.org

Textile Waste Categories and Trends

Typical patterns of recycling and reuse in textile waste management are summarized in the following table.

Material	Availability	Reuse Trend	Recycling Trend
Cotton	High	Increasing	Moderate
Polyester	Medium	Moderate	Stable
Nylon	Medium	Moderate	Moderate
Polyester Mix	Medium	Moderate	Low

4.5.2 Feature Engineering

Raw marketplace and waste creation data are transformed into features appropriate for predictive analytics models through feature engineering.

Primary Features

Feature	Description
material_type	Type of textile waste
historical_waste_quantity	Previous waste quantities generated
listing_frequency	Number of marketplace listings over time
geographic_location	Location of textile waste generation
time_period	Month or seasonal period

These characteristics aid in determining patterns of trash creation both geographically and temporally.

Derived Features and Target Variables

The reuse and recycling potential will be estimated using derived indicators.

Feature	Formula	Purpose
Predicted Waste Volume	Historical_quantity × Growth_rate	Estimate future waste generation
Recyclable Waste Quantity	Total_waste × (Recycling_rate / 100)	Estimate recycling potential
Reusable Waste Quantity	Total_waste × (Reuse_rate / 100)	Estimate reuse potential
Remaining Waste Quantity	Total_waste − (Recyclable + Reusable)	Estimate disposal volume
Marketplace Availability	Total_listed − Sold_quantity	Determine current market supply

Data-driven decision-making for the management of textile waste is made possible by these factors.

4.5.3 Model Development and Selection

Two methods of analysis will be assessed.

Method A: Baseline Statistical Trend Analysis

To comprehend past trends, conventional statistical methods will be applied.

Techniques consist of:

- Moving Average Analysis to reduce volatility.
- Visualizing time series to spot long-term patterns.
- Analyzing seasonal patterns to identify cyclical waste production.

These methods offer a standard against which machine learning models may be compared.

Method B: Forecasting using Machine Learning

Based on past trends, machine learning algorithms will forecast future amounts of textile waste.

Among the potential models are:

1. **Regression in Linear Form**
   To find linear correlations between waste volumes and time, a straightforward prediction model was employed.
2. **Regressor for Random Forests**
   An ensemble algorithm that can recognize non-linear correlations among many features.
3. **XGBoost, or Gradient Boosting Regressor**
   On structured datasets, this potent boosting method frequently yields excellent predicted accuracy.

Input Features for Forecasting Model

Feature	Description
Historical waste quantities	Past material waste data
Material types	Textile category
Listing frequency	Marketplace activity level
Geographic location	Source region
Time variable	Month or seasonal period

Future waste quantities for various textile materials will be forecasted by the model's output.

4.5.4 Dashboard and Report Generation

To display statistics and forecasts related to textile waste, an interactive analytics dashboard will be created.

Features of the Dashboard

The dashboard will consist of:

- Visualization of accessible textile waste materials in real time
- Trend graphs illustrating the production of trash over time.
- Confidence interval-based forecast graphics.
- Maps showing the geographic distribution of the sources of textile waste.
- Important sustainability metrics include:
    - rate of reuse
    - rate of recycling
    - total amount of garbage kept out of landfills.

Generation of Reports

For stakeholders, the technology will automatically provide analytical reports.

Reporting features consist of:

- Reports that can be exported in CSV or PDF
- Report generation on a scheduled basis.
- Report parameters that can be customized include:
    - ranges of dates
    - sorts of materials
    - geographical places

These studies aid in the strategic decision-making process for programs aimed at recycling and reusing textile waste.

4.5.5 Marketplace Platform Development

Textile waste products may be efficiently traded and redistributed thanks to the digital marketplace platform.

Front-end Programming

The following will be used to create the user interface:

- Using React.js to create dynamic webpages.
- A set of Material-UI components for responsive design.
- Mobile-friendly design for cross-platform accessibility.

Features of the Platform.

The market will offer:

- Role-based access control for user registration and authentication (manufacturers, purchasers, administrators)
- AI-assisted categorization for waste listing production (connected with Module 1)
- Advanced filtering and searching by:
    - Sort of material
    - Location
    - Quality grade
    - Cost
- Environmental effect measurements are displayed on product description pages (Module 2).
- Information on traceability with digital material passports (Module 3)
- Processing of transactions between buyers and sellers
- Customized dashboards for users

4.5.6 API Development

The backend services will be implemented as RESTful APIs using FastAPI.

API Endpoints

Endpoint	Description
GET /api/waste-forecast	Retrieve predicted waste quantities.
GET /api/marketplace/listings	Browse available textile waste listings.
POST /api/marketplace/listing	Create a new waste listing.
GET /api/analytics/reports	Generate analytical reports

4.7 API Development and Integration (All Modules)

FastAPI will be used to construct each of the four modules as a separate microservice, guaranteeing independent scalability and loose coupling.

Typical API Features:

- JSON request/response in a RESTful design
- Pydantic models for input validation
- Complete error management with educational messages
- Automated documentation for OpenAPI (Swagger)
- Rate limitation to safeguard APIs.
- JWT-based verification.

Architecture of Integration:

- API Gateway forwards requests to the relevant microservices
- RabbitMQ/Kafka message queue for asynchronous communication
- Finding services to handle dynamic endpoints.
- Centralized monitoring and logging.

Deployment and Containerization:

- Docker containers for every microservice to guarantee uniformity in the environment.
- For local development, use Docker Compose
- Kubernetes for orchestration in production (if necessary)
- AWS, GCP, and Azure cloud deployment.

Testing Methodology:

- Unit tests for specific functions
- Tests of API endpoint integration
- Validating performance through load testing
- End-to-end testing for the whole process.

5. High-level system architecture

5.1 Overview of Proposed Solution

The suggested system is a digital marketplace platform driven by artificial intelligence that is intended to make it easier for Sri Lanka's clothing sector to effectively redistribute, reuse, and upcycle textile waste. Together, the platform's four clever modules convert unstructured textile waste into standardized digital assets that are enhanced with data on price, traceability, environmental effect, and categorization.

Each module may function independently while interacting via a centralized API layer thanks to the system architecture's microservices-based design. Scalability, maintainability, and integration flexibility are all enhanced by this design.

The platform's web-based marketplace interface allows users to upload textile waste, analyze it using AI models, and display it for possible purchasers. The backend services process photos, track the lifespan of textile waste materials, estimate resale values, compute environmental benefits, and provide suggestions.

As a result, the system promotes data-driven decision-making and openness in the handling of textile waste, supporting circular economy principles.

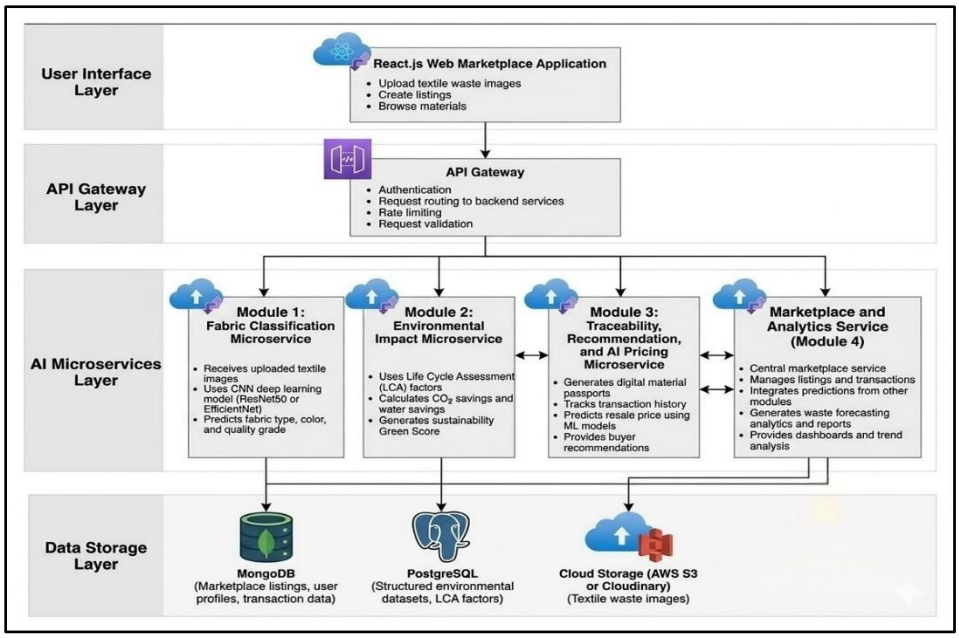

Figure 6: Conceptual System Architecture.

5.2 System Components

Module 1: AI Fabric Classification

Goal: Using uploaded photos, automatically determine the features of textile waste.

Important Elements

- Handler for uploading images.
- Module for preparing images (normalization, resizing)
- CNN-based deep learning model (either EfficientNet with transfer learning or ResNet50)
- Module for post-processing predictions
- FastAPI endpoint: /classify_fabric

For every image of textile waste, the module predicts the fabric type, colour, and quality grade.

Module 2: Environmental Impact Prediction

Goal: Calculate the sustainability advantages of reusing textile waste.

Important Elements

- Database of Lifecycle Assessment (LCA) factors
- Pipeline for feature engineering.
- Prediction models based on regression (Random Forest, XGBoost)
- Calculator for Green Score Sustainability
- Endpoint for FastAPI: /predict_impact

The module forecasts:

- Decrease in CO2 emissions.
- Water conservation
- Each material batch's sustainability score.

Module 3: Traceability, Recommendations & AI Pricing

Goal: Make intelligent marketplace matching and lifecycle monitoring possible.

Important Elements

- Passport generator for digital materials
- Ledger of transactions for traceability
- AI-powered price forecasting system
- Buyer-material matching recommendation engine.
- Endpoints for FastAPI: /predict_price, /get_recommendations, and /traceability/{batch_id}
- Transparency, equitable pricing, and effective redistribution of textile waste are guaranteed by this module.

Module 4: Marketplace Platform & Predictive Analytics

Goal: Offer the main analytics and user interface features.

Important Elements

- Front-end marketplace application using React.js.
- Role management and user authentication
- Management of textile waste listings
- An engine for searching and filtering.
- Prediction model for waste forecasting
- Dashboard for interactive analytics
- Module for generating reports.

To facilitate real-time marketplace operations and decision-making, this module combines the outputs from the AI services.

Data Storage Architecture

Main Database: MongoDB

Used for adaptable document-based storage, such as:

- Listings of textile waste
- Profiles of users
- Transaction histories
- Passports made of digital materials.

Relational Information System

PostgreSQL

Used with structured datasets such as:

- Lifecycle assessment (LCA) elements
- Categories of textile materials
- Coefficients of environmental effect

Storage of Files

Cloud-based storage solutions like:

- S3 on AWS
- Cloudinary

The AI classification algorithm uses these services to store uploaded photos of textile waste.

5.3 Component Interactions and Data Flow

An explanation of data flow

Waste Uploaded by the Seller:

- The seller uses the marketplace interface to take and post garbage images.
- Sending an image to the Module 1 API for categorization
- Module 1 provides confidence scores along with fabric type, color, and quality grade.
- The marketplace shows AI recommendations, which the seller verifies or fixes.

Calculating the Environmental Impact:

- Verified characteristics provided to the Module 2 API
- Module 2 generates a Green Score, computes CO2/water savings, and obtains LCA factors.
- Results were stored along with a waste listing.

Cost and Suggestions:

- Sending waste characteristics to the Module 3 API
- Module 3 uses past data to forecast the ideal price.
- Potential buyers and reuse options are found by recommendation engines.
- Results are displayed and returned to the vendor.

Creation of Traceability:

- Module 3 creates a digital material passport with a distinct batch ID.
- Database-stored passport; QR code created for listing.
- Every subsequent purchase was recorded on the passport.

Listing on the Marketplace:

- All AI-generated data was used to construct a comprehensive waste listing.
- Buyers with search and filter capabilities can view the listing.
- Displayed environmental parameters and traceability information.

Interaction with Buyers:

- Buyers look through and search listings.
- View comprehensive product pages with pricing, traceability, and a green score.
- Make a purchase or get in touch with the seller.

Analytics for prediction:

- Module 4 periodically examines past waste data.
- Future waste volumes by material type are predicted by a forecasting model.
- The analytics dashboard shows trends and forecasts.
- Reports produced for decision-making by stakeholders.

5.4 Scalability and Security Considerations

Scalability

- The microservices design allows modules to scale independently in response to demand.
- Orchestration (Kubernetes) and containerization (Docker) for autonomous scaling.
- Distributing the load among several high-demand service instances.
- Performance-enhancing database indexing and query optimization.
- CDN for the distribution of static assets.

Safety:

- All API queries are authenticated using JWT.
- Role-based access control (admin, buyer, and manufacturer)
- Encrypting all data in transit via HTTPS
- Encrypting databases to protect sensitive data.
- Validating and sanitizing inputs to stop injection attacks.
- Limiting rates to stop API misuse.
- Frequent changes to dependencies and security audits.

6. Requirements

6.1 Stakeholder Identification

Table 7: Stakeholder Identification.

Stakeholder	Description	Key Needs
Manufacturers (Sellers)	Textile factories and garment manufacturers that generate fabric waste during production processes.	Efficient waste listing, fair pricing recommendations, additional revenue generation from waste, and sustainability reporting metrics.
Designers / Upcyclers (Buyers)	Independent designers, artisans, and small-scale creators seeking affordable textile materials for reuse and upcycling.	Access to affordable raw materials, reliable quality information, clear material specifications, and environmental impact data.
SMEs	Small and medium enterprises involved in sustainable fashion and textile reuse initiatives.	Reliable material supply, traceability of textile sources, cost predictability, and easy marketplace access.
Recycling Companies	Organizations that process textile waste into recycled fibers or new materials.	Consistent waste supply streams, detailed material composition data, and forecasting of available waste volumes.
Platform Administrators	System managers are responsible for operating and maintaining the digital marketplace platform.	User management tools, transaction monitoring, system analytics, fraud detection, and platform maintenance capabilities.
Sustainability Officers	Corporate or organizational ESG (Environmental, Social, and Governance) stakeholders are responsible for sustainability initiatives.	Environmental impact metrics, automated sustainability reporting, and tracking of circular economy progress.
Policy Makers	Government agencies and industry regulators oversee environmental	Access to industry insights, textile waste reduction metrics, and data-driven support for policy development.

	compliance and waste management policies.	

6.2 Functional Requirements

Table 8: Functional Requirements.

ID	Requirement Description	Module
FR-01	The system shall accept textile waste image uploads in common formats such as JPEG and PNG.	M1
FR-02	The system shall analyze uploaded images and predict the fabric type, color, and quality grade using a trained CNN model.	M1
FR-03	The system shall return confidence scores ranging from 0.0 to 1.0 for each predicted attribute.	M1
FR-04	The system shall maintain a database containing Life Cycle Assessment (LCA) factors for at least 15 different textile materials.	M2
FR-05	The system shall calculate the estimated carbon dioxide (CO_2) emissions saved based on material type, weight, and LCA factors.	M2
FR-06	The system shall calculate the estimated water savings associated with the reuse or recycling of textile waste.	M2
FR-07	The system shall generate a composite sustainability indicator ("Green Score") based on combined CO_2 and water savings.	M2
FR-08	The system shall allow users to input textile waste information, including material type, quantity, quality grade, condition, and images.	M3, M4
FR-09	The system shall predict an optimal resale price for textile waste using an AI-based pricing model.	M3
FR-10	The system shall provide intelligent recommendations for potential reuse, recycling, or upcycling opportunities.	M3
FR-11	The system shall record material information, transactions, and recommendation outcomes in the traceability module.	M3

FR-12	The system shall generate digital material passports with unique batch identifiers for tracking textile waste.	M3
FR-13	The system shall provide a dashboard displaying textile waste trends and marketplace analytics.	M4
FR-14	The system shall analyze historical waste data to forecast future textile waste volumes.	M4
FR-15	The system shall allow users to filter and analyze marketplace data by material type, quantity, location, and time period.	M4
FR-16	The system shall generate analytical reports based on predicted waste volumes and marketplace data.	M4
FR-17	The system shall support user registration, login, and role-based authentication for different stakeholders.	M4
FR-18	The system shall allow buyers to search, filter, and view available textile waste listings.	M4
FR-19	The system shall display environmental impact metrics such as CO_2 savings and water savings on product detail pages.	M2, M4
FR-20	The system shall expose core functionality through RESTful APIs with proper documentation.	All
FR-21	The system shall handle invalid inputs and errors gracefully and return informative error messages to users.	All

6.3 Non-Functional Requirements

Table 9: Non-Functional Requirements

ID	Requirement Description	Target
NFR-01	Accuracy of the fabric classification model	Greater than **85% accuracy** on the test dataset
NFR-02	Accuracy of environmental impact prediction models	**R^2 greater than 0.85**
NFR-03	API response time for prediction requests	**Less than 2 seconds** (95th percentile)
NFR-04	Dashboard data processing performance	**Less than 5 seconds** for standard datasets
NFR-05	System scalability for concurrent users	Tested with up to **50 simultaneous users**
NFR-06	System scalability for future growth	Ability to support additional materials and users without major architectural changes
NFR-07	System reliability and availability	**99% uptime** during development and testing
NFR-08	Usability of the user interface	Interface should be intuitive for **non-technical users**, validated through user testing.
NFR-09	Interpretability of sustainability metrics	The **Green Score** should be easily understandable for non-expert users
NFR-10	Maintainability of the system codebase	Modular architecture with proper documentation and version control (Git)
NFR-11	Maintainability of environmental data	The LCA factor database should be easily updatable as new data becomes available.

NFR-12	System security and access control	Role-based authentication and secure data transmission
NFR-13	Protection of sensitive waste data	Prevention of unauthorized access through authentication and access control mechanisms
NFR-14	Robustness of the classification model	The model should maintain **>70% accuracy** under varied lighting conditions and image angles.
NFR-15	Transparency of system outputs	System responses should include **data source citations** and explanations where applicable.

6.4 System Requirements

A cutting-edge technological stack that facilitates cloud deployment, scalable web apps, and artificial intelligence development will be used to create the suggested system. Effective model training, dependable backend services, and a responsive user interface are guaranteed by the frameworks and tools chosen.

Table 10: System Requirements

Component	Technology / Requirement
Programming Languages	Python 3.9+ for backend and machine learning development, JavaScript for frontend development
Web Frameworks	FastAPI for backend microservices and REST API development, React.js for frontend user interface
Machine Learning Libraries	TensorFlow 2.x / Keras for deep learning models, scikit-learn for regression models, pandas and numpy for data processing
Databases	MongoDB for flexible document storage, PostgreSQL for structured relational data, Redis for caching and session management
Cloud Platforms	Cloud hosting platforms such as Amazon Web Services (AWS), Google Cloud Platform, or Render
Containerization	Docker and Docker Compose for consistent deployment and environment management.
Version Control	Git for source code management with repositories hosted on GitHub or GitLab
API Communication	RESTful APIs using HTTP/HTTPS protocols
Development Tools	Visual Studio Code for development and Jupyter Notebooks for machine learning experimentation
Testing Tools	pytest for backend testing, Jest for frontend testing, and Postman for API testing

1. User Requirements

For Manufacturers/Sellers:

- Having access to a camera or smartphone to take pictures of garbage.
- Basic knowledge of the waste items in the list
- Waste can be placed on a clear, contrasting background for optimal categorization outcomes.
- AI or computer vision expertise is not necessary.

For Purchasers (Upcyclers/Designers):

- Basic knowledge of sustainability indicators (water use, carbon footprint)
- Capacity to analyze environmental indicators and the Green Score
- Knowledge of material kinds and quality ratings

For Integrators/Platform Developers:

- Access to base URLs and API docs.
- Knowledge of integrating REST APIs.
- Credentials for authentication to access APIs.

6.5 Use Cases

Table 11: Use Case 1 - Upload Waste Image for Classification

Element	Description
Use Case ID	UC-01
Use Case Name	Upload Waste Image for Automated Classification
Summary	A seller uploads a textile waste image, and the system automatically analyzes the image using an AI classification model to suggest fabric attributes for the marketplace listing.
Priority	High
Primary Actor(s)	Seller (Manufacturer)
Secondary Actor(s)	Marketplace Platform, Fabric Classification Microservice
Preconditions	The seller must be logged into the platform and must be on the "Create New Listing" page.
Postconditions	The system displays AI-generated predictions for fabric type, color, and quality grade with confidence scores, and the fields are pre-filled in the listing form.
Trigger	The seller clicks the **"Upload Image"** button and selects a textile waste image file.
Main Scenario	1. The seller selects the image file. 2. Platform sends image to /classify fabric endpoint. 3. Microservice validates, pre-processes, and runs inference. 4. Returns JSON with predictions and confidence scores. 5. Platform pre-fills form fields. 6. Seller reviews and proceeds.
Extensions	• E1 (Low Confidence): If confidence <0.7, highlight the field for manual review • E2 (Upload Error): Display error message if API fails
Open Issues	• Determining the optimal confidence threshold for automated suggestions.

	• Handling images containing multiple textile pieces or mixed materials.

Table 12: Use Case 2 - Generate Environmental Impact Prediction

Element	Description
Use Case ID	UC-02
Use Case Name	Generate Environmental Impact Prediction
Summary	The system automatically calculates and returns environmental impact metrics for the waste batch.
Priority	High
Primary Actor(s)	Main Marketplace Platform (System)
Secondary Actor(s)	None
Preconditions	Main platform received listing request; fabric type, weight, and quality grade provided
Postconditions	Impact metrics (CO_2 saved, water saved, Green Score) are calculated and returned.
Trigger	The platform sends a POST request to /predict_impact upon a new listing.
Main Scenario	1. Platform sends a request with fabric type, weight, and quality. 2. Microservice validates input. 3. Retrieves corresponding LCA factors. 4. Calculates total CO_2 saved and water saved. 5. Calculates composite Green Score 6. Returns JSON response with HTTP 200
Extensions	• E1 (Invalid Input): Return error if fabric type not found. • E2 (Missing Input): Return 422 Unprocessable Entity
Open Issues	• Handling blended fabrics (weighted average calculation)

Table 13: Use Case 3 - Seller Uploads Waste for Pricing and Traceability

Element	Description
Use Case ID	UC-03
Name	Seller Uploads Textile Waste for Pricing and Traceability
Summary	Seller uploads textile waste information to the platform. The system processes the data to generate AI-based pricing predictions, reuse or recycling recommendations, and traceability records.
Priority	High
Primary Actor(s)	Seller / Manufacturer
Secondary Actor(s)	Backend System (AI Pricing Model, Recommendation Engine, Traceability System)
Preconditions	Seller must be registered and logged into the platform; required textile waste details must be available.
Postconditions	Textile waste information is stored in the system with traceability data, and the seller can view AI-generated pricing, recommendations, and lifecycle tracking information.
Trigger	Seller navigates to the **"Upload Textile Waste"** section of the platform.
Main Scenario	1. Seller enters textile waste details (fabric type, weight, quality grade, condition, images). 2. The platform sends the data to the backend via API. 3. Backend validates the input information. 4. AI pricing model predicts the optimal resale price. 5. The recommendation engine generates reuse/recycling suggestions. 6. The traceability system logs the material information and transaction data. 7. The platform displays results including predicted price, recommendations, and traceability report. 8. Seller reviews the results and decides whether to publish the listing.
Extensions	• **E1 – Incomplete Data:** System prompts the seller to correct missing information.

	• **E2 – Processing Error:** Error message displayed and retry option provided. • **E3 – Duplicate Material Entry:** System warns the seller to avoid duplicate listings.
Open Issues	Scalability testing for larger datasets and user loads; continuous evaluation of AI pricing accuracy; standardization of traceability data formats.

Table 14: Use Case 4 - View Waste Forecast Dashboard

Element	Description
Use Case ID	UC-04
Name	View Waste Forecast Dashboard
Summary	Buyers, sellers, and stakeholders view visualized trends and predicted waste quantities for informed decision-making.
Priority	High
Primary Actor(s)	Buyers, Sellers, Sustainability Stakeholders
Secondary Actor(s)	System, Analytics Module, Database
Preconditions	User registered and logged in; historical waste data available; analytics module active.
Postconditions	Dashboard displays updated waste trends and forecasts; users can filter data.
Trigger	User wants to analyze historical data and view predictions.
Main Scenario	1. User logs in 2. Navigates to the waste forecasting dashboard. 3. System retrieves historical waste data. 4. Analytics module processes data, generates trends, and forecasts. 5. Dashboard displays visual charts, tables, and key insights. 6. User applies filters, downloads report as needed.
Extensions	**E1** (Data Unavailable): System shows "no data" message. **E2** (Forecasting Error): System displays error notification.

Open Issues	Integrating predictive analytics for mixed-material waste forecasting

6.6 Wireframes

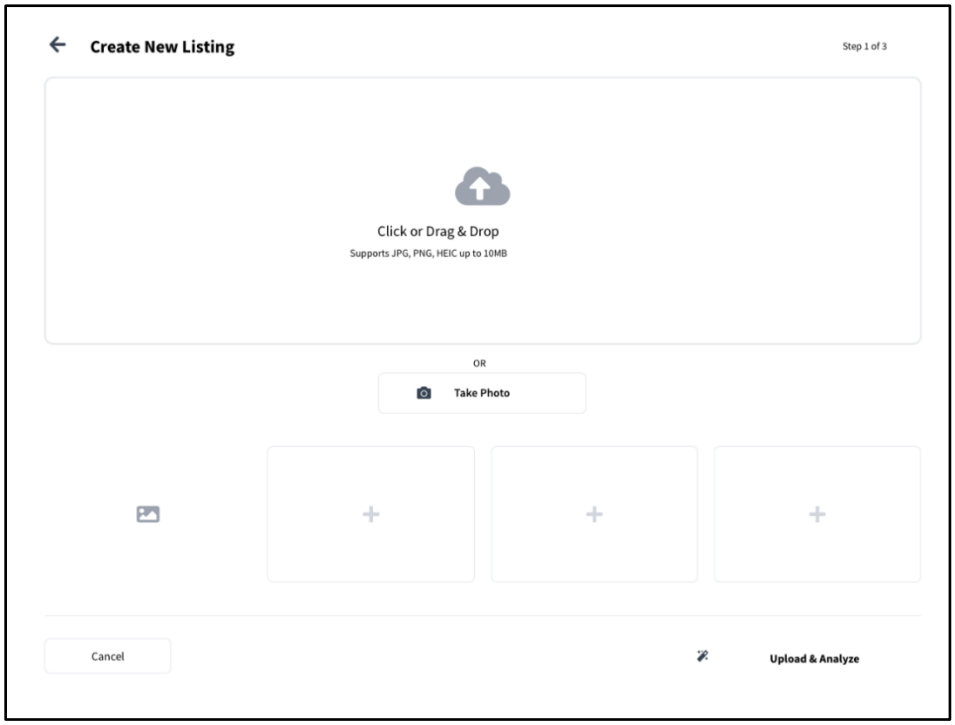

Figure 7: Wireframe - Image Upload Interface for Sellers

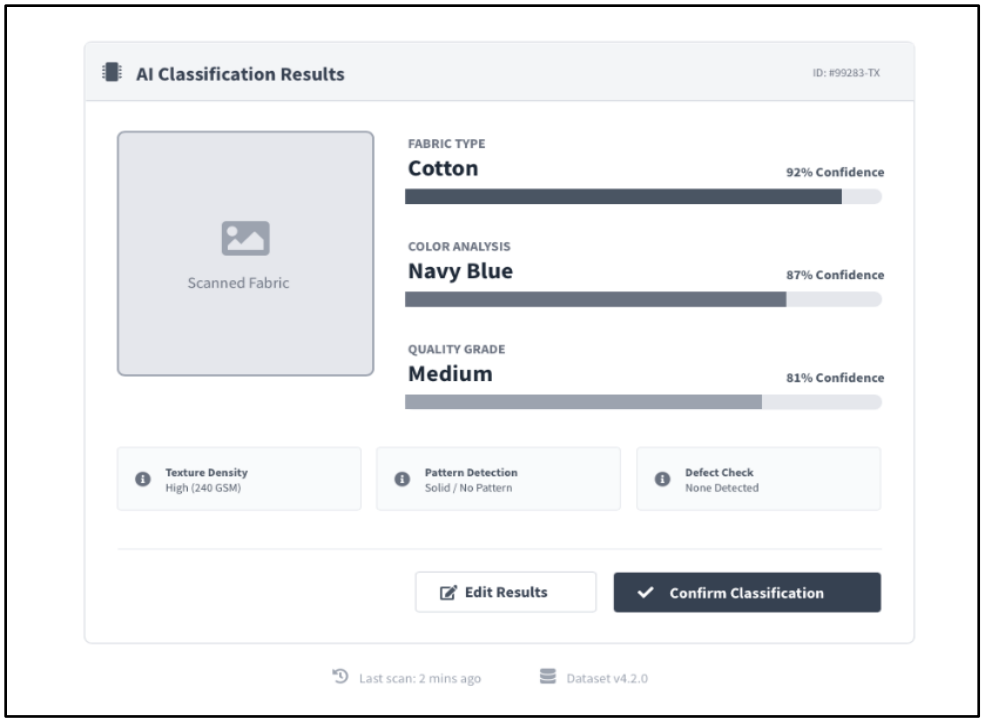

Figure 8: Wireframe - Classification Results Display

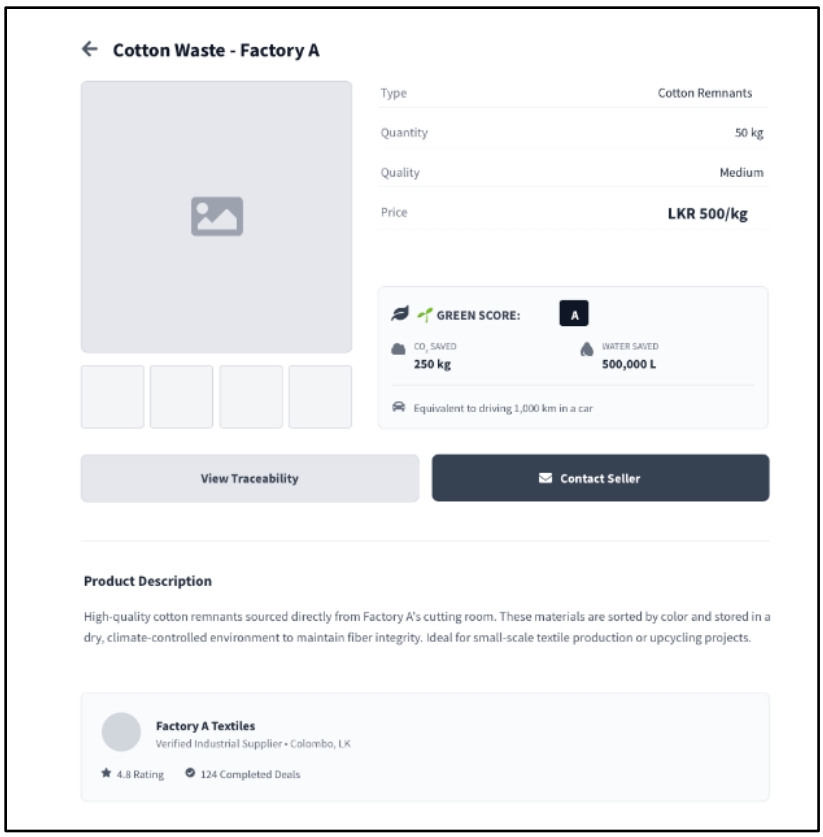

Figure 9: Wireframe - Product Listing with Green Score

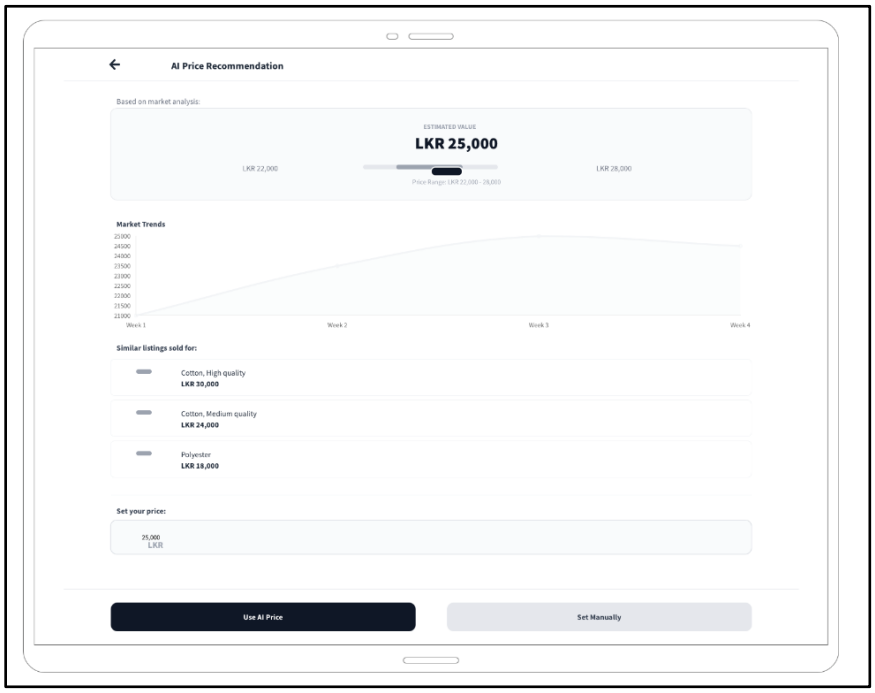

Figure 10: Wireframe - AI Pricing Display

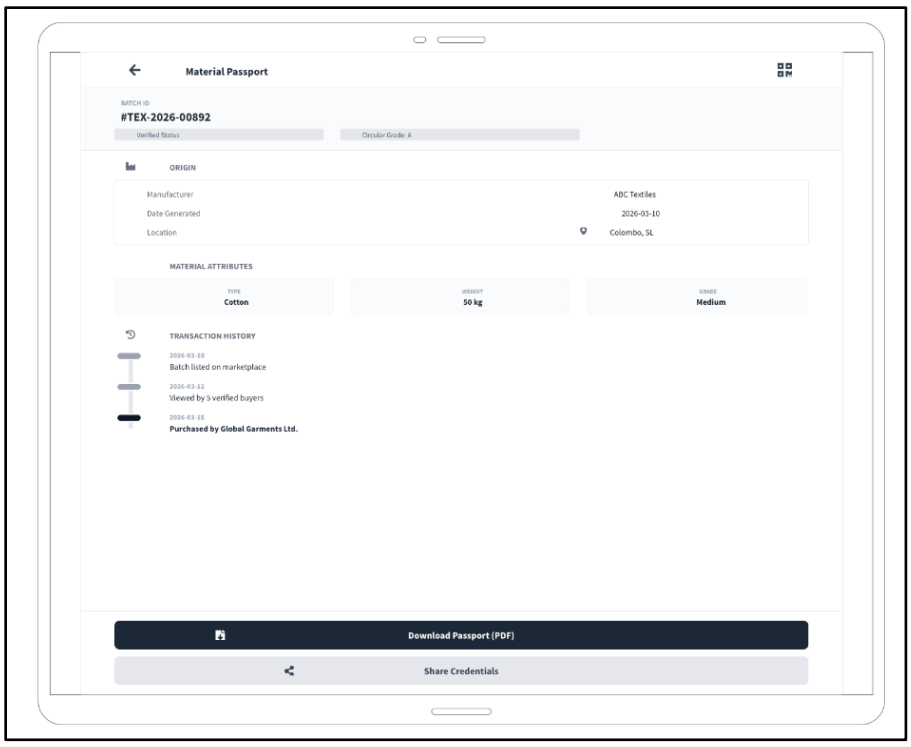

Figure 11: Wireframe - Traceability Information

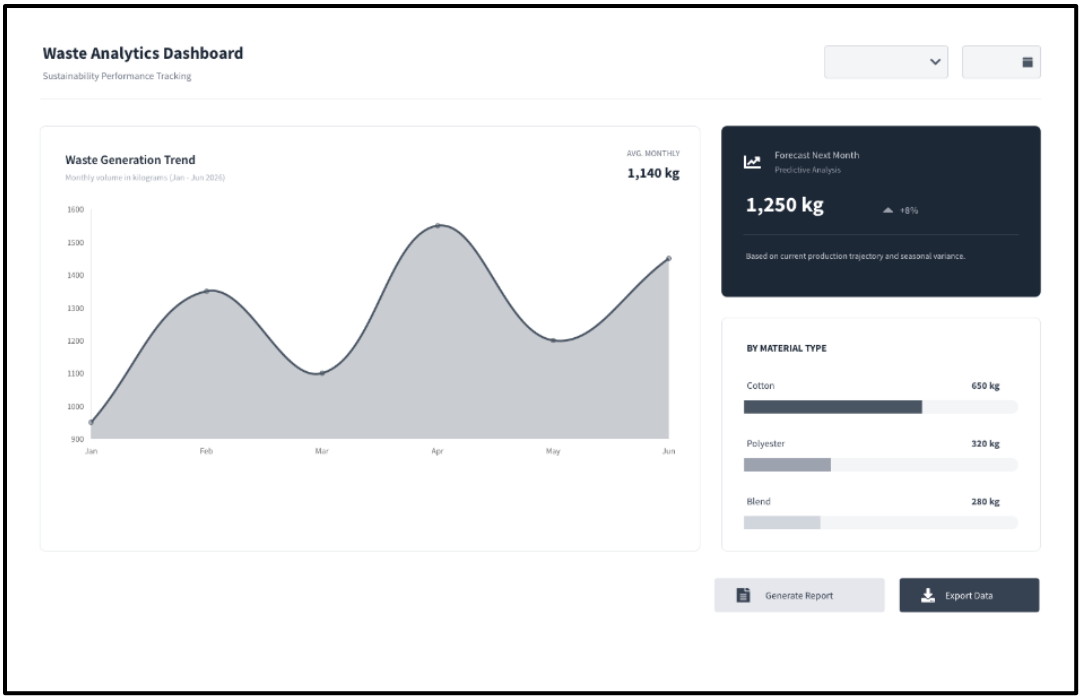

Figure 12: Wireframe - Waste Forecasting Dashboard

7. Commercialization plan

7.1 Target Market

Primary Market:

- Textile Manufacturers & Producers: Businesses that produce a lot of textile waste are looking for effective ways to dispose of it and possible sources of income.
- Designers and Upcyclers: Small to medium-sized companies that prioritize sustainable fashion and need dependable suppliers of raw materials.
- SMEs in the Circular Economy: Companies that focus on trading sustainable materials, upcycling, and recycling textiles.

Secondary Market:

- Corporate Sustainability Teams: Companies in need of sustainability analytics and ESG reporting.
- Waste Management Companies: Businesses looking for an automated inventory and sorting system.
- Facilities that need pre-sorted textile waste streams are known as recycling plants.
- Research Institutions: Establishments that focus on sustainable textiles and the circular economy.

Market Space: The global market for artificial intelligence (AI) in smart recycling and waste management is expanding rapidly. With potential for regional development, this module focuses on the South Asian textile and apparel value chain, especially Sri Lanka's clothing sector.

7.2 Customer Personas

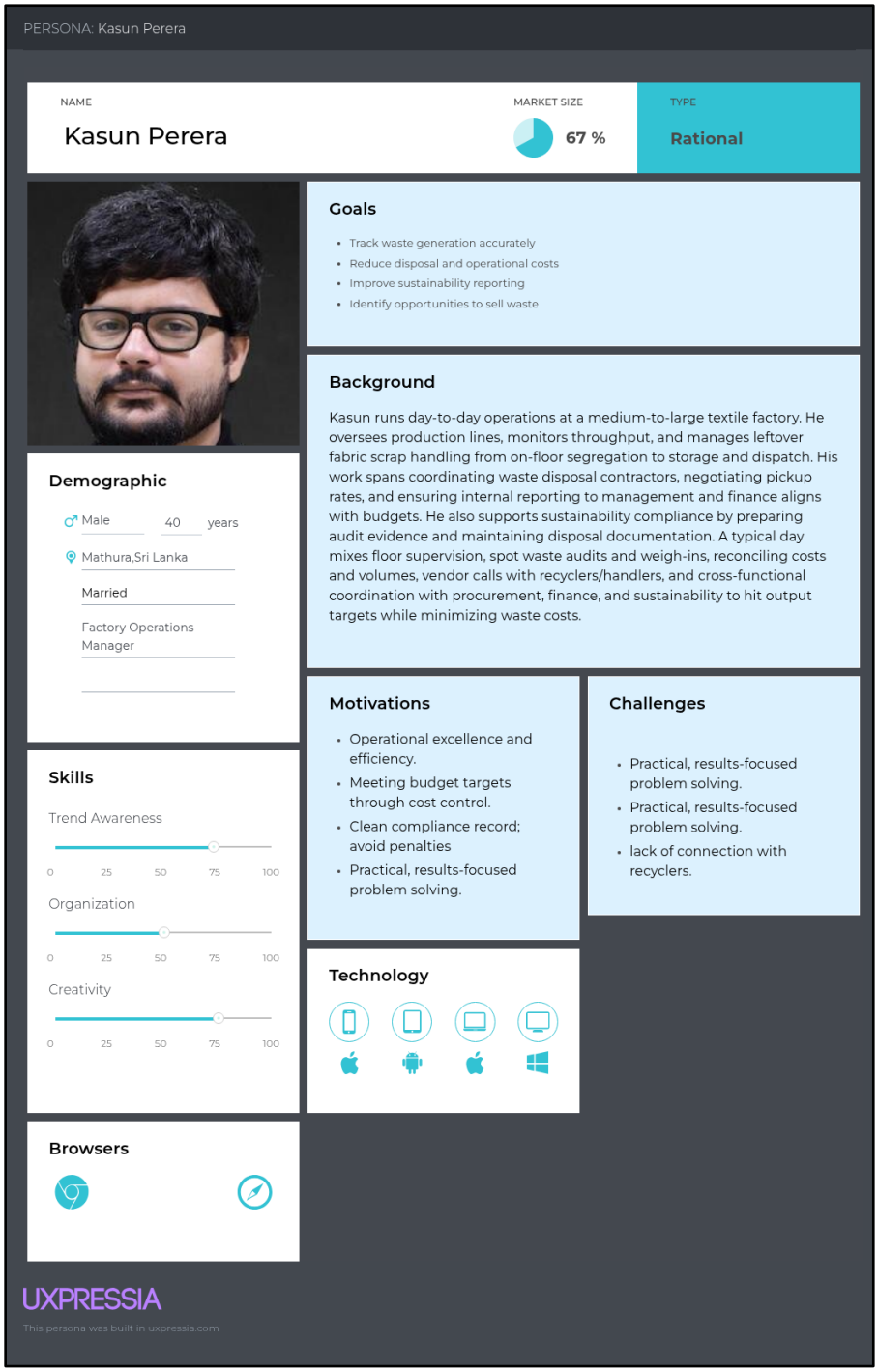

Figure 13: Persona 1 - Factory Manager

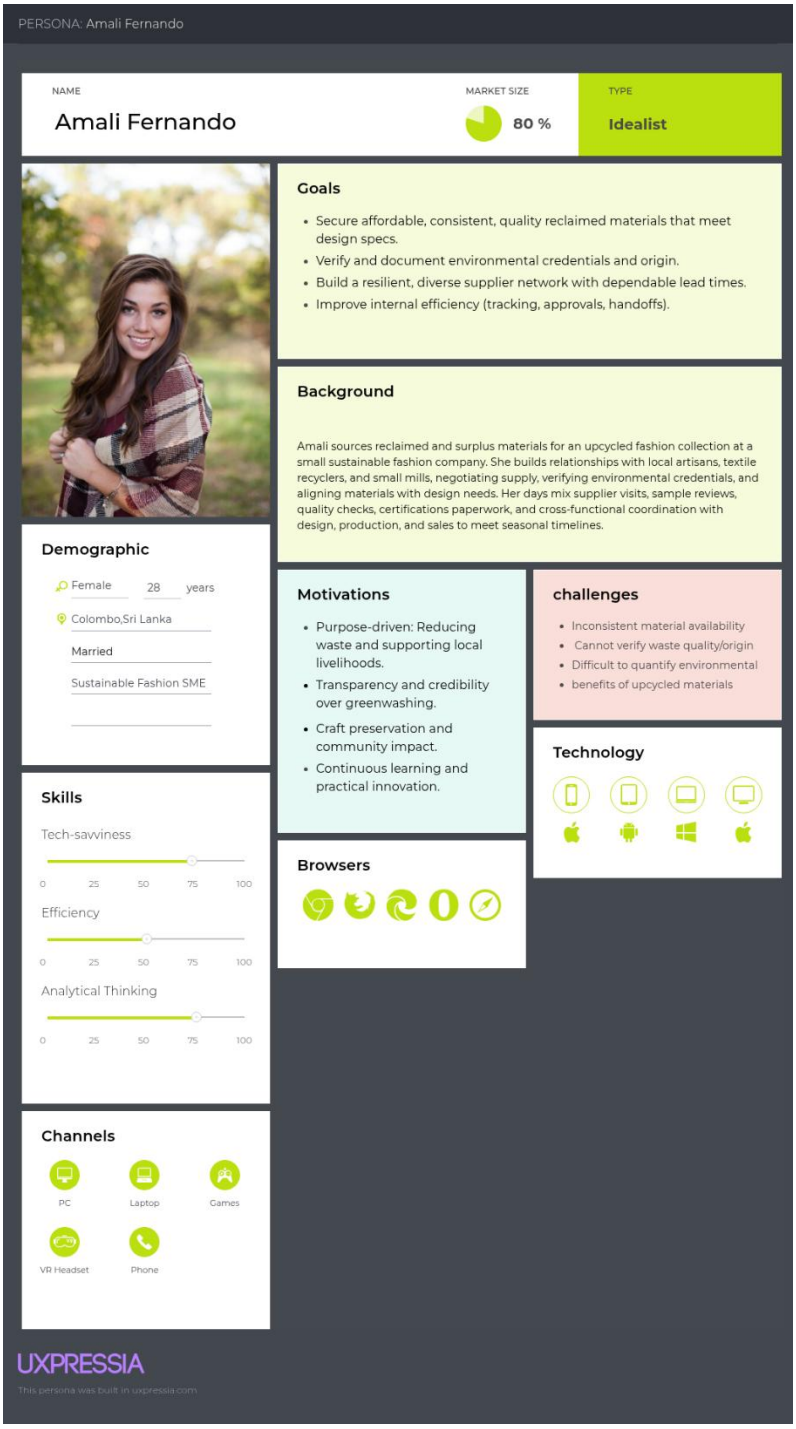

Figure 14: Persona 2 - Eco-conscious Textile Manager

7.3 Value Proposition

For Producers (Vendors):

- Time Savings: Reduces human labor by automating trash listing through AI categorization.
- Revenue Generation: Verified environmental credentials allow for "green premium"; AI pricing determines the best market value.
- ESG Reporting: Sustainability reporting is enhanced by traceability and impact measurements.
- Waste Reduction: Planning waste management solutions is aided by predictive analytics.

For Purchasers (Upcyclers/Designers):

- Material Access: A dependable supplier of reasonably priced, quality-checked raw materials
- Making Informed Decisions: Sustainable purchasing is supported by environmental effect data (Green Score).
- Transparency: Total traceability increases confidence in the quality and provenance of materials.
- Verified sustainability certifications for final products are a marketing advantage.

Regarding the Platform:

- Market Differentiation: Compared to generic markets, integrated AI capabilities.
- Data Network Effects: Expanding transaction databases increases the precision of AI models.
- Ecosystem Position: Creates a platform as a pioneer in textile circular economy technologies.

7.4 Revenue Model

1. Model Based on Subscriptions:
   - Basic: Limited listings per month, free marketplace access
   - Professional: AI categorization, limitless listings, and pricing suggestions ($20/month)
   - Enterprise: $100/month for custom reports, advanced analytics, and API access
2. Model of Transaction Fees:
   - Fee depending on a percentage of successful transactions (5–10%)
3. Subscription to the API:
   - Pay-per-call APIs for impact prediction, pricing, or categorization.
   - Monthly subscription tiers for high use
4. High-end Reports:
   - Analytical research and industry insights purchased once.
5. Advisory Services:
   - Advice on the circular economy, training, and custom integration

7.5 Cost Estimation

Table 15: Cost Estimation Monthly.

Cost Category	**Estimated Monthly (LKR)**
Cloud Infrastructure	LKR **15,500 – 31,000**
Database Hosting	LKR **6,200 – 9,300**
API Gateway / Load Balancing	LKR **3,100 – 6,200**
Development Tools	**LKR 0** (open source)
Marketing	LKR **15,500 – 31,000** (initial)
Total Monthly	**LKR 40,300 – 77,500**

Student research covers the initial development expenditures; additional funding will be required for scalability.

1. Strategy for Pricing

Model of Freemium

- The Free Tier
- Cost: LKR 0 per month

Five postings every month

- Simple search
- Typical reports
- Tiers of Payment
- Infinite listings
- AI characteristics
- High-level analytics
- Priority assistance

Table 16: Tiered Pricing.

Plan	Price (USD)	Price (LKR)	Features
Basic	$0 / month	**LKR 0 / month**	Essential marketplace access
Professional	$20 / month	**LKR 6,200 / month**	Full AI features, recommendations, 100 listings
Enterprise	$100 / month	**LKR 31,000 / month**	Unlimited listings, API access, custom reports, dedicated support

Competitive Advantage

Competitor Type	Limitation	Our Advantage
Generic B2B Marketplaces	No textile specialization and no AI-driven insights	Domain-specific AI designed for textile waste and circular economy applications
Manual Waste Exchanges	Inefficient processes and inconsistent data handling	Automated waste classification and standardized data management
Basic Recycling Platforms	Lack of pricing intelligence and market analytics	AI-powered waste valuation and recommendation systems
Traditional LCA Consultants	Expensive services, slow analysis, and limited accessibility	Automated system providing instant environmental impact calculations

Important Differentiators

1. AI Integrated Throughout the Whole Process

   AI is integrated into every stage of the waste management lifecycle, including pricing, matching, categorization, and sustainability analysis.

2. Textile Waste Dataset Particular to Sri Lanka

   The accuracy of machine learning models is improved for the Sri Lankan sector by incorporating locally relevant textile waste data.

3. Extensive Intelligence on Sustainability

   The platform offers comprehensive information on:

   - Impact on the environment
   - Economic worth
   - Traceability of materials

4. Architecture of Microservices

   Scalable deployment, flexibility, and simpler system changes are made possible by a modular microservices design.

5. Emphasis on the Circular Economy

   Instead of just enabling transactions, the platform places a higher priority on resource reuse, waste reduction, and sustainable value generation.

7.8 Intellectual Property Considerations

- Innovative CNN Model Architecture: Tailored for the categorization of textile waste.
- Curated Textile Trash Dataset: Post-industrial trash photos from Sri Lanka
- LCA Integration Methodology: An innovative method for quantifying effect at the batch level
- Platform Codebase: Implementation of a full-stack marketplace
- Branding: It is possible to trademark a platform's name, logo, and interface design.
- Options for Licensing: API licensing to other platforms to generate extra income.

8. Budget and justification

Table 17: Project Budget Breakdown

Component	Amount (USD)	Amount (LKR)	Justification
Data Collection and Materials			
Fabric Waste Samples	20.00	6,000	Obtaining diverse waste samples from local tailors and manufacturers for dataset creation
Photography Setup	10.00	3,000	Basic lighting and background setup for consistent image capture
Cloud Computing Resources			
Cloud GPU Credits (Training)	30.00	9,000	CNN model training and hyperparameter tuning
Cloud Hosting for APIs (4 months)	40.00	12,000	Hosting microservices during testing and demonstration
Database Hosting	15.00	4,500	MongoDB / PostgreSQL instances
Infrastructure and Tools			
Development Tools	0.00	0	All tools are open-source (Python, TensorFlow, VS Code, Git)
Internet Charges (Research)	20.00	6,000	Research activities, dataset downloads, and cloud access over 5 months
Knowledge and Consultation			
Technical Consultation (AI/ML)	15.00	4,500	Guidance on model architecture and validation
Industry Consultation (Textile)	15.00	4,500	Validation of classification categories and LCA assumptions

Sustainability Expert	10.00	3,000	Green Score methodology validation
Contingency			
Miscellaneous Expenses	15.00	4,500	Unforeseen costs, additional samples, and extended cloud usage
TOTAL	**190.00**	**57,000**	Estimated total across all four modules

Remarks:

- Four team members each received a budget for the demands of their own modules.
- When feasible, free cloud service tiers will be used.
- Budget items are supplemented by university resources (lab computers, library access).
- Total conservative estimate: optimization may result in lower real costs

9. Work breakdown structure.

Table 18: Work Breakdown Structure

WBS ID	Task / Component	Description	Assigned To
1.0	**Project Management**	**Overall coordination and integration**	**All**
1.1	Project Planning	Scope definition, milestone setting	All
1.2	Progress Tracking	Weekly meetings, status reports	All
1.3	Integration Management	Module coordination, API alignment	All
1.4	Final Documentation	Consolidated report preparation	All
2.0	**Module 1: AI Fabric Classification**	**Module development**	**Sandeepa**
2.1.1	Literature Review	CNNs, textile classification studies	Sandeepa
2.1.2	Define Categories & Rubric	Final category list, labeling guidelines	Sandeepa
2.1.3	Set up Development Environment	Python, TensorFlow, cloud GPU	Sandeepa
2.2.1	Source Fabric Waste Samples	Collect from local tailors/manufacturers	Sandeepa
2.2.2	Capture Images	1,500–2,000 varied condition images	Sandeepa
2.2.3	Label Images	Type, color, quality labeling	Sandeepa
2.2.4	Implement Data Augmentation	Rotation, flips, brightness	Sandeepa
2.2.5	Create Train/Validation/Test Splits	70/15/15 division	Sandeepa
2.3.1	Implement Baseline CNN	Simple ConvNet baseline	Sandeepa
2.3.2	Implement Transfer Learning	ResNet50/EfficientNet	Sandeepa
2.3.3	Train Initial Model	Frozen base layers	Sandeepa
2.3.4	Fine-tune Model	Unfrozen layers, low learning rate	Sandeepa
2.3.5	Hyperparameter Tuning	Optimize for accuracy	Sandeepa
2.4.1	Evaluate Validation Set	Monitor overfitting	Sandeepa

2.4.2	Final Test Evaluation	Accuracy, precision, recall, F1	Sandeepa
2.4.3	Error Analysis	Qualitative review	Sandeepa
2.5.1	Set up FastAPI Project	Project structure	Sandeepa
2.5.2	Implement /classify_fabric	Inference endpoint	Sandeepa
2.5.3	Integrate Trained Model	Model loading, prediction logic	Sandeepa
2.5.4	Validation & Error Handling	Input validation, exceptions	Sandeepa
2.5.5	Write Unit/Integration Tests	Test coverage	Sandeepa
2.6.1	Docker Containerization	Dockerfile creation	Sandeepa
2.6.2	Deploy to Cloud VM	Running container	Sandeepa
2.6.3	Document API Usage	Swagger/OpenAPI	Sandeepa
3.0	**Module 2: Environmental Impact Prediction**	**Module development**	**Siriwardhana**
3.1.1	LCA Literature Review	Textile environmental impacts	Siriwardhana
3.1.2	Identify Materials & Factors	Cotton, polyester, etc.	Siriwardhana
3.1.3	Document Requirements	System specifications	Siriwardhana
3.2.1	Source LCA Data	Ecoinvent, literature, Higg MSI	Siriwardhana
3.2.2	Create Structured Dataset	CSV with material factors	Siriwardhana
3.2.3	Synthetic Training Data	Vary weights and grades	Siriwardhana
3.3.1	Baseline Lookup Model	Rule-based calculation	Siriwardhana
3.3.2	Linear Regression Model	Simple baseline	Siriwardhana
3.3.3	Random Forest Regressor	Ensemble method	Siriwardhana
3.3.4	Gradient Boosting (XGBoost)	Advanced algorithm	Siriwardhana
3.4.1	Data Split	Train/test division	Siriwardhana
3.4.2	Train Models	Model training	Siriwardhana
3.4.3	Evaluate Performance	R^2, MAE, RMSE	Siriwardhana
3.4.4	Tune Best Model	Optimization	Siriwardhana
3.5.1	Set up FastAPI Project	Project structure	Siriwardhana
3.5.2	Implement /predict_impact	Impact prediction endpoint	Siriwardhana
3.5.3	Implement Green Score	Composite sustainability score	Siriwardhana
3.5.4	Validation & Error Handling	Input validation	Siriwardhana

3.5.5	Write Tests	Unit/integration tests	Siriwardhana
3.6.1	Docker Containerization	Docker packaging	Siriwardhana
3.6.2	Deploy Microservice	Cloud deployment	Siriwardhana
3.6.3	Document API	API documentation	Siriwardhana
4.0	**Module 3: Traceability & AI Pricing**	**Module development**	**Madushanka**
4.1.1	Literature Review	Traceability, recommendation systems	Madushanka
4.1.2	Define Data Requirements	Attributes, historical data	Madushanka
4.1.3	Document Specifications	System requirements	Madushanka
4.2.1	Gather Waste Attributes	Fabric type, weight, quality	Madushanka
4.2.2	Collect Historical Transactions	Past sales data	Madushanka
4.2.3	Create Dataset	Structured CSV dataset	Madushanka
4.3.1	Baseline Pricing Model	Simple calculation	Madushanka
4.3.2	Linear Regression Pricing	Pricing baseline	Madushanka
4.3.3	Random Forest Pricing	Advanced pricing model	Madushanka
4.3.4	Recommendation Engine	Content/collaborative filtering	Madushanka
4.4.1	Digital Passport Schema	Batch attributes and history	Madushanka
4.4.2	Ledger System	Transaction logging	Madushanka
4.4.3	QR Code Generator	Batch traceability access	Madushanka
4.5.1	Set up FastAPI Project	Project structure	Madushanka
4.5.2	Implement /predict_price	Pricing endpoint	Madushanka
4.5.3	Implement /get_recommendations	Recommendation endpoint	Madushanka
4.5.4	Implement /traceability/{id}	Passport retrieval	Madushanka
4.5.5	Write Tests	Unit/integration tests	Madushanka
4.6.1	Docker Containerization	Docker packaging	Madushanka
4.6.2	Deploy Microservice	Cloud deployment	Madushanka
4.6.3	Document API	API documentation	Madushanka
5.0	**Module 4: Marketplace Platform**	**Platform development**	**Thalagala**
5.1.1	Literature Review	Digital platforms, forecasting	Thalagala
5.1.2	Define User Requirements	Stakeholder needs	Thalagala

5.1.3	System Architecture Design	Component design	Thalagala
5.2.1	Gather Waste Data	Type, quantity, location	Thalagala
5.2.2	Collect Historical Records	Waste trends	Thalagala
5.2.3	Structure Dataset	Forecasting dataset	Thalagala
5.3.1	Analyze Historical Data	Trend identification	Thalagala
5.3.2	Baseline Forecasting	Statistical forecasting	Thalagala
5.3.3	ML Forecasting Model	Random Forest/XGBoost	Thalagala
5.3.4	Validate Forecasting Model	Accuracy assessment	Thalagala
5.4.1	Dashboard Layout Design	Visualization planning	Thalagala
5.4.2	Interactive Charts	Real-time analytics	Thalagala
5.4.3	Report Generator	PDF/CSV export	Thalagala
5.5.1	Set up React Project	Frontend structure	Thalagala
5.5.2	User Authentication	Registration, login, roles	Thalagala
5.5.3	Listing Interface	Waste upload forms	Thalagala
5.5.4	Search & Filters	Marketplace browsing	Thalagala
5.5.5	API Integration	Connect all modules	Thalagala
5.6.1	Deploy Frontend	Static hosting/CDN	Thalagala
5.6.2	Deploy Backend APIs	Cloud deployment	Thalagala
5.6.3	End-to-End Testing	Workflow validation	Thalagala
5.6.4	Platform Documentation	User & technical guides	Thalagala
6.0	**System Integration**	**Connect all modules**	**All**
6.1	Performance Testing	Load and stress testing	All
6.2	Security Audit	Vulnerability assessment	All
6.3	Final Report Compilation	Complete documentation	All
6.4	Presentation Preparation	Demo system and slides	All

10. Gantt chart

Task ID	Task Description	Start Week	End Week	Duration (Weeks)
Phase 1: Research & Planning (Weeks 1–4)				
1.1	Literature Review – All Modules	1	3	3
1.2	Define Requirements & Specifications	2	4	3
1.3	Set up Development Environments	3	4	2
Phase 2: Module 1 – Fabric Classification				
2.1	Source Fabric Samples	4	6	3
2.2	Capture & Label Images	5	9	5
2.3	Implement Data Augmentation	8	9	2
2.4	Develop Baseline CNN	9	11	3
2.5	Implement Transfer Learning	10	13	4
2.6	Model Training & Tuning	11	15	5
2.7	Model Evaluation	14	16	3
2.8	API Development	16	19	4
2.9	Deployment & Documentation	19	22	4
Phase 3: Module 2 – Environmental Impact				
3.1	Source LCA Data	4	6	3
3.2	Create LCA Factor Dataset	6	8	3
3.3	Feature Engineering	8	10	3
3.4	Develop Baseline Model	9	11	3
3.5	Implement ML Models	10	13	4
3.6	Model Training & Validation	12	15	4
3.7	API Development	15	18	4
3.8	Green Score Implementation	16	19	4
3.9	Deployment & Documentation	18	22	5
Phase 4: Module 3 – Traceability & Recommendations				
4.1	Data Collection	4	7	4
4.2	Feature Engineering	7	9	3

4.3	Develop Pricing Models	8	12	5
4.4	Develop Recommendation Engine	10	14	5
4.5	Design Traceability Framework	12	15	4
4.6	Implement Digital Passport	14	17	4
4.7	API Development	16	20	5
4.8	Deployment & Documentation	19	22	4
Phase 5: Module 4 – Marketplace & Analytics				
5.1	Data Collection	4	7	4
5.2	Develop Forecasting Models	7	12	6
5.3	Design Dashboard & Reports	10	14	5
5.4	Develop Frontend (React)	12	18	7
5.5	Implement User Authentication	14	16	3
5.6	Create Listing Interface	16	19	4
5.7	API Integration	18	21	4
5.8	Deployment & Documentation	20	23	4
Phase 6: Integration & Final Deliverables				
6.1	System Integration Testing	20	22	3
6.2	Performance & Security Testing	21	23	3
6.3	Final Report Compilation	22	24	3
6.4	Presentation Preparation	23	24	2

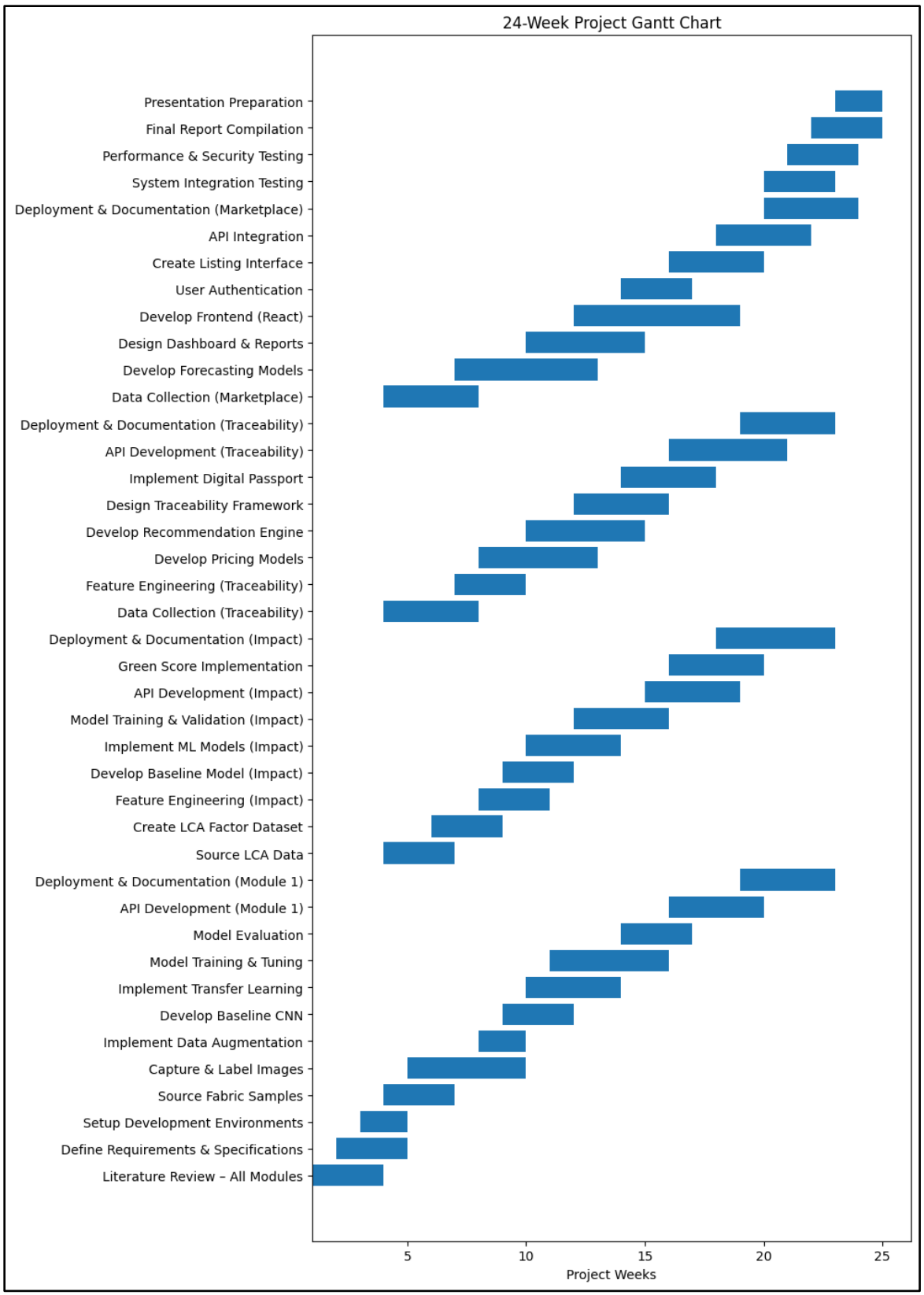

Figure 15: Gantt Chart

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12. Appendices

12.1 Risk Analysis

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy
R1	Insufficient dataset size for CNN training	Medium	High	Apply data augmentation, use transfer learning, and generate synthetic data to expand the dataset.
R2	Low model accuracy (<85%)	Medium	High	Perform iterative tuning, experiment with ensemble methods, and test multiple architectures.
R3	LCA data availability limitations	Low	Medium	Use multiple authoritative sources and clearly document assumptions
R4	API integration challenges between modules	Medium	Medium	Define clear API contracts early and perform regular integration testing
R5	Cloud cost overruns	Low	Medium	Use free-tier services, monitor usage, and optimize resource allocation
R6	Schedule delays due to system complexity	Medium	High	Follow Agile methodology, prioritize core features, and include buffer time.
R7	Team member availability issues	Low	High	Maintain regular communication, cross-train members, and maintain clear documentation.
R8	Model overfitting	Medium	Medium	Apply cross-validation, regularization techniques, and maintain a separate test dataset.
R9	Data quality issues (noisy labels)	Medium	High	Create a clear labeling rubric, perform multiple verifications, and consult experts.

R10	Technology stack compatibility issues	Low	Medium	Research compatibility early and use stable framework versions

12.2 Ethical Considerations

The following ethical guidelines are followed in this study:

1. **Data Confidentiality and Privacy:**

- No personally identifying information will be gathered from people.
- If manufacturer information is supplied, it will be anonymized.
- Data protection laws are followed in all data storage.

2. **Knowledgeable Consent:**

- Participants who supply fabric samples will be made aware of the study's objectives.
- Participation is voluntary with the option to withdraw.

3. **Transparency and Accuracy of Data:**

- Every source of LCA data will be properly referenced.
- Limitations of the model will be recorded.
- Users will be made aware that AI forecasts are approximations rather than assurances.

4. **Ecological Accountability:**

- The goal of research is to lessen the damage that textile waste does to the environment.
- Encourages sustainable forms of production and consumption.

5. **Honesty in Academics:**

- Every source is correctly referenced in IEEE format.
- Declared original work, no plagiarism.
- Reproducibility made possible via a transparent technique.

6. **The principle of no harm**

- Sensitive information or human beings are not used in research.
- Participants did not suffer any physical or psychological damage.
- Minimal hazards are outweighed by benefits to society and the environment.

12.3 System Architecture Diagram

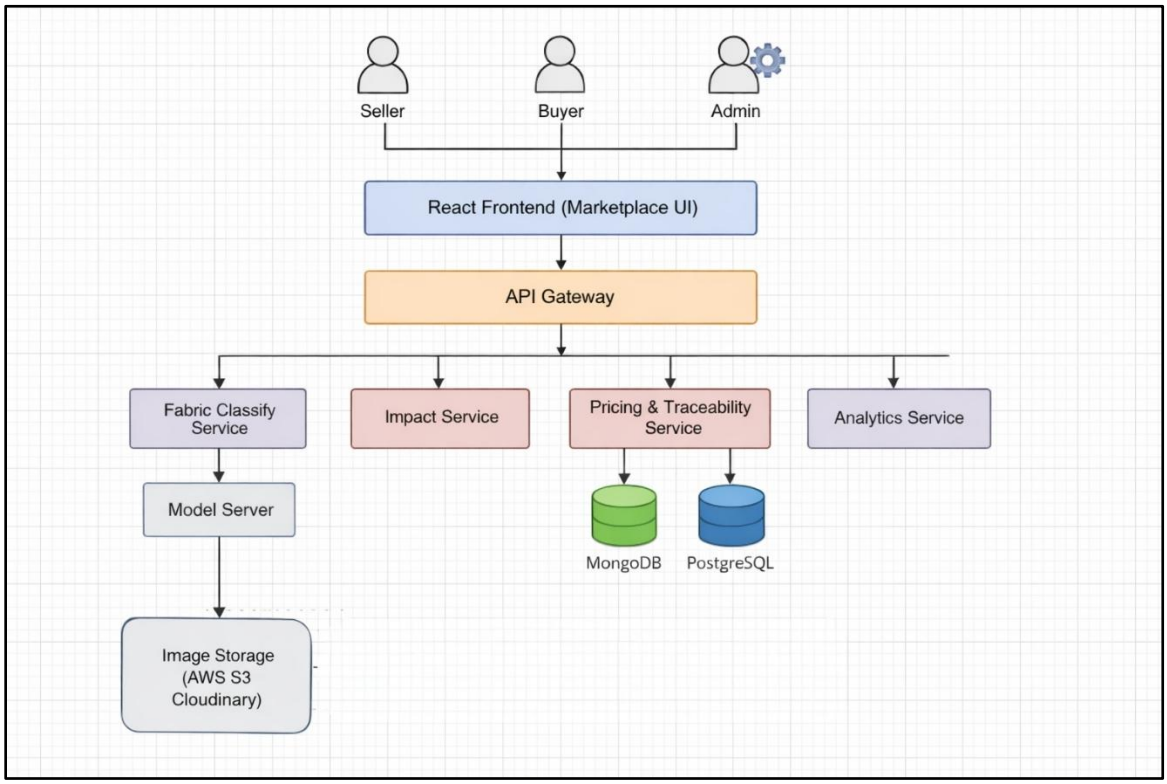

Figure 16: System Architecture Diagram

12.4 UML Class Diagram

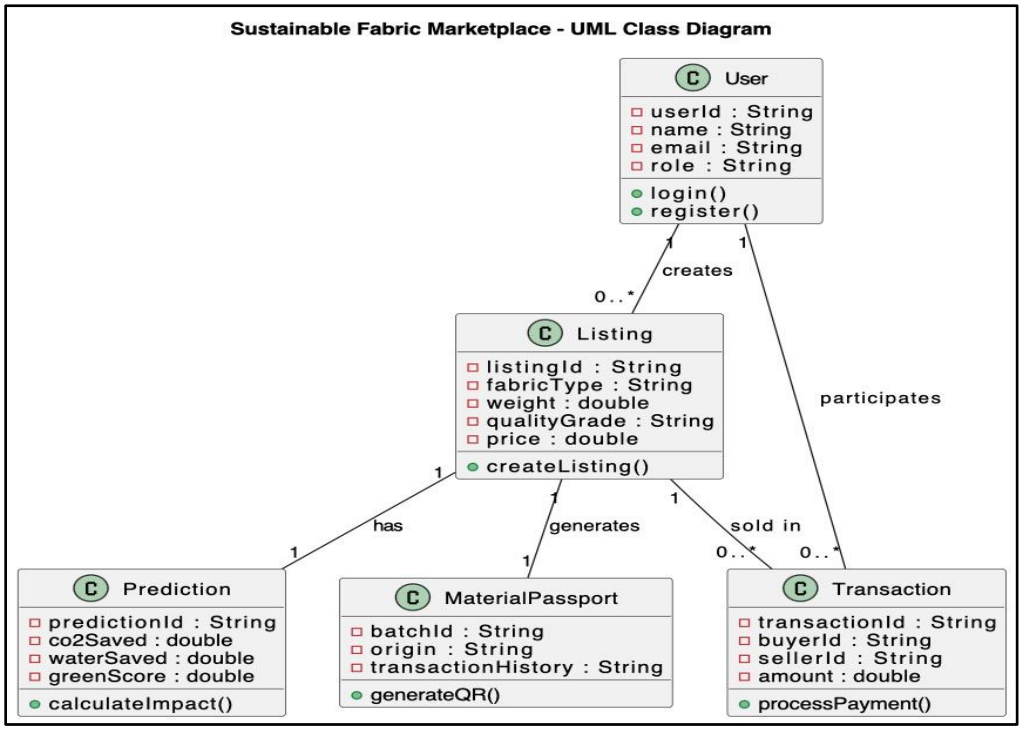

Figure 17: UML Class Diagram.

12.5 Sequence Diagram

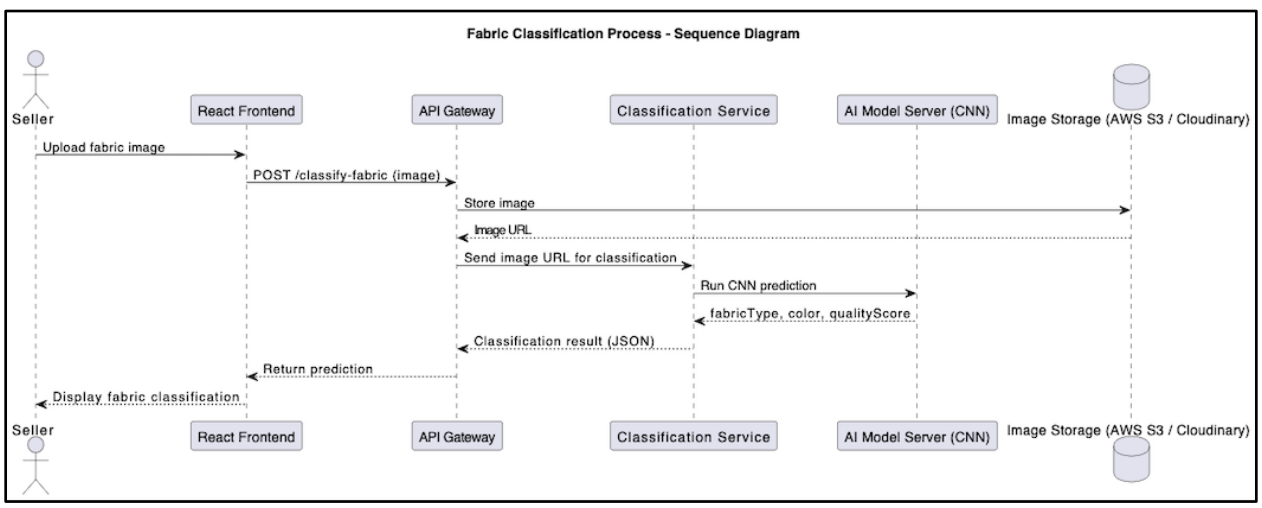

Figure 18: Sequence Diagram

12.6 Database Schema Diagram

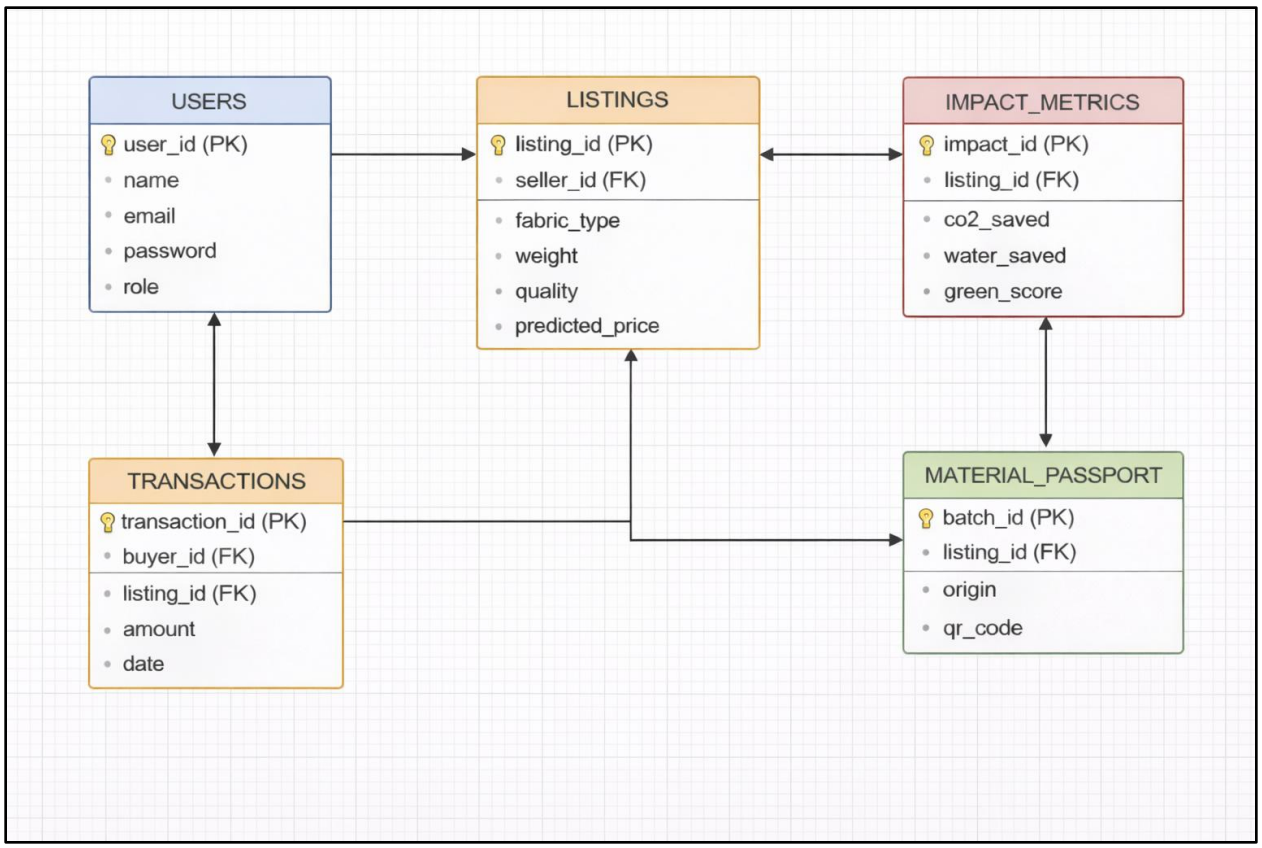

Figure 19: Database Schema Diagram

12.7 Deployment Architecture Diagram

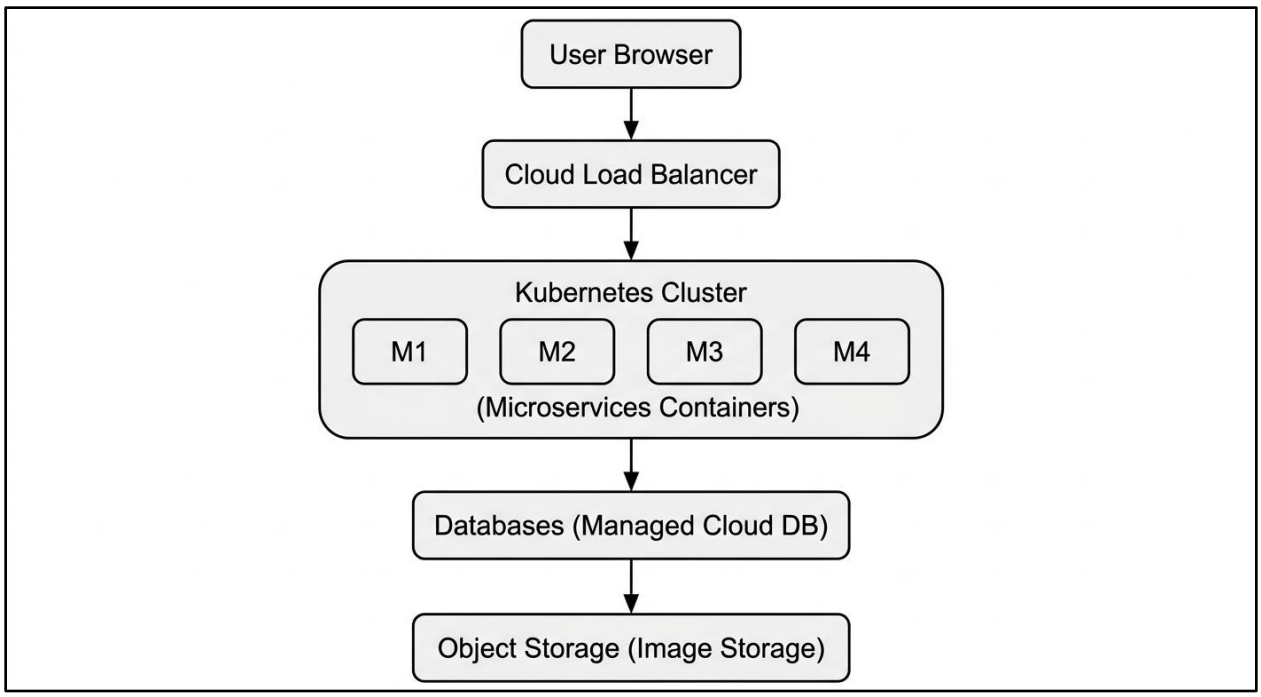

Figure 20: Deployment Architecture Diagram